

FRC Driver Station Powered by NI LabVIEW / 由 NI LabVIEW 提供支持的 FRC Driver Station

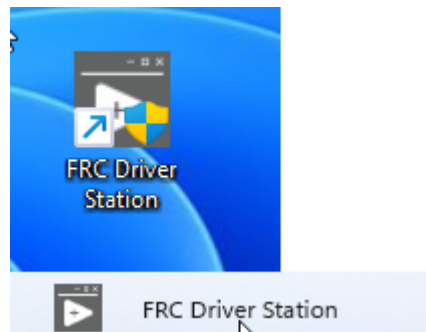
This article describes the use and features of the FRC® Driver Station Powered by NI LabVIEW.

本文介绍由 NI LabVIEW 提供支持的 FRC® Driver Station 的用法和功能。

For information on installing the Driver Station software see the Installing the FRC Game Tools document in the WPILib documentation.

有关安装 Driver Station 软件的信息，请参阅 WPILib 文档中的“Installing the FRC Game Tools”。

Starting the FRC Driver Station / 启动 FRC Driver Station



The FRC Driver Station can be launched by double-clicking the icon on the Desktop or by selecting Start->All Apps->FRC Driver Station.

可以双击 Desktop 上的图标，或依次选择 Start->All Apps->FRC Driver Station，启动 FRC Driver Station。

NOTE: The FRC Driver Station can be configured on the [Setup Tab](#) to launch SmartDashboard and Shuffleboard. WPILib must be installed to use SmartDashboard and Shuffleboard.

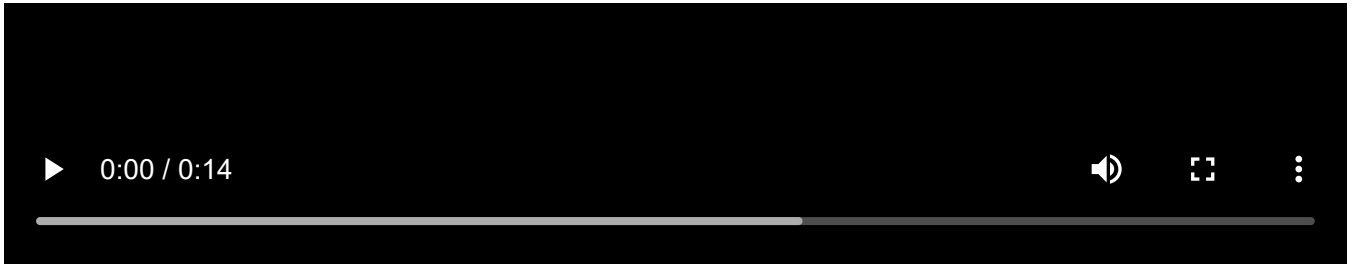
说明：可以在 [Setup Tab](#) 中配置 FRC Driver Station，使其启动 SmartDashboard 或 Shuffleboard。必须安装 WPILib 才能使用 SmartDashboard 和 Shuffleboard。

Driver Station Key Shortcuts / Driver Station 键盘快捷键

- **F1** - Force a Joystick refresh.
F1——强制刷新 Joystick。
- **[+] + ** - Enable the robot (the 3 keys above Enter on most keyboards)
**[+] + **——启用机器人（在大多数键盘上，这三个按键位于 Enter 上方）。
- **Enter** - Disable the Robot
Enter——禁用机器人。
- **Space** - Emergency Stop the robot. After an emergency stop is triggered the roboRIO will need to be rebooted before the robot can be enabled again.
space——紧急停止机器人。触发 Emergency Stop 后，必须重新启动 roboRIO，之后才能再次启用机器人。
- **Backspace** - "A-Stop" the robot when in Practice Mode - Autonomous Enabled. The robot will be

disabled until the Practice Mode reaches Teleop then will be automatically re-enabled. While disabled, the application background will flash orange indicating the robot will re-enable automatically.

Backspace ——当机器人处于 Practice Mode - Autonomous Enabled 时对其执行“A-Stop”。机器人将保持禁用，直到 Practice Mode 进入 Teleop，随后自动重新启用。机器人处于禁用状态时，应用程序背景会闪烁橙色，表示机器人将自动重新启用。



图片提醒：此处应插入图片。图片描述：Driver Station flashing orange in A-Stop mode / Driver Station 在 A-Stop 模式下闪烁橙色；原图片路径：`images/driver-station/ds-astop.png`。

View this page in the web documentation for an animation of the A-Stop behavior

在 Web 文档中查看此页面，可以看到 A-Stop 行为的动画。

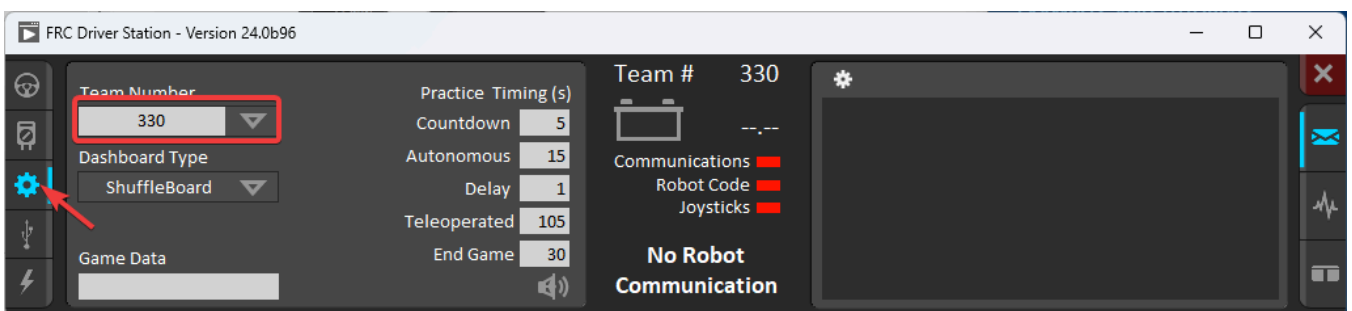
NOTE: Space bar will E-Stop the robot regardless of if the Driver Station window has focus or not

说明：无论 Driver Station 窗口是否具有焦点，按下 Space Bar 都会对机器人执行 E-Stop。

WARNING: When connected to FMS in a match, teams must press the Team Station E-Stop button to emergency stop their robot as the DS enable/disable and E-Stop key shortcuts are ignored.

警告：比赛中连接到 FMS 时，DS 的启用/禁用和 E-Stop 键盘快捷键会被忽略，因此队伍必须按下 Team Station E-Stop Button 才能紧急停止机器人。

Setting Up the Driver Station / 设置 Driver Station



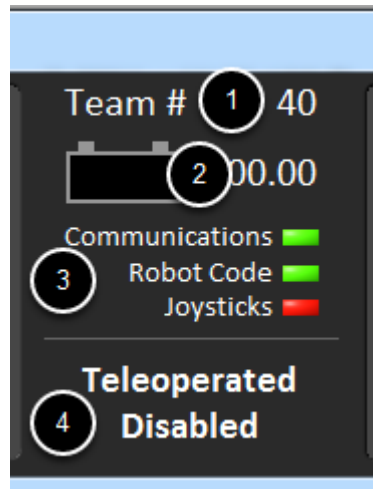
The DS should be set to your team number in order to connect to your robot. In order to do this click the Setup tab then enter your team number in the team number box. Press return or click outside the box for the setting to take effect.

为了连接到机器人，应将 DS 设置为队伍编号。为此，请单击 Setup Tab，然后在队伍编号输入框中输入队伍编号。按 Return，或单击输入框以外的位置，使设置生效。

PCs will typically have the correct network settings for the DS to connect to the robot already, but if not, make sure your Network adapter is set to DHCP.

PC 通常已经具备使 DS 能够连接到机器人的正确网络设置；如果没有，请确认 Network Adapter 已设置为 DHCP。

Status Pane / Status Pane



The Status Pane of the Driver Station is located in the center of the display and is always visible regardless of the tab selected. It displays a selection of critical information about the state of the DS and robot:

Driver Station 的 Status Pane 位于显示区域中央，无论选择哪个 Tab 都始终可见。它显示关于 DS 和机器人状态的若干关键信息：

1. Team # - The Team number the DS is currently configured for. This should match your FRC team number. To change the team number see the Setup Tab.

Team #——DS 当前配置的队伍编号。它应与 FRC 队伍编号一致。若要更改队伍编号，请参阅 Setup Tab。

2. Battery Voltage - If the DS is connected and communicating with the roboRIO this displays current battery voltage as a number and with a small chart of voltage over time in the battery icon. The background of the numeric indicator will turn red when the roboRIO brownout is triggered. See the roboRIO Brownouts document for more information.

Battery Voltage——如果 DS 已连接并正在与 roboRIO 通信，此处会以数值形式显示当前电池电压，并在电池图标中显示一张电压随时间变化的小型图表。触发 roboRIO 欠压时，数值指示器的背景会变为红色。有关更多信息，请参阅 WPILib 文档中的“roboRIO Brownouts”。

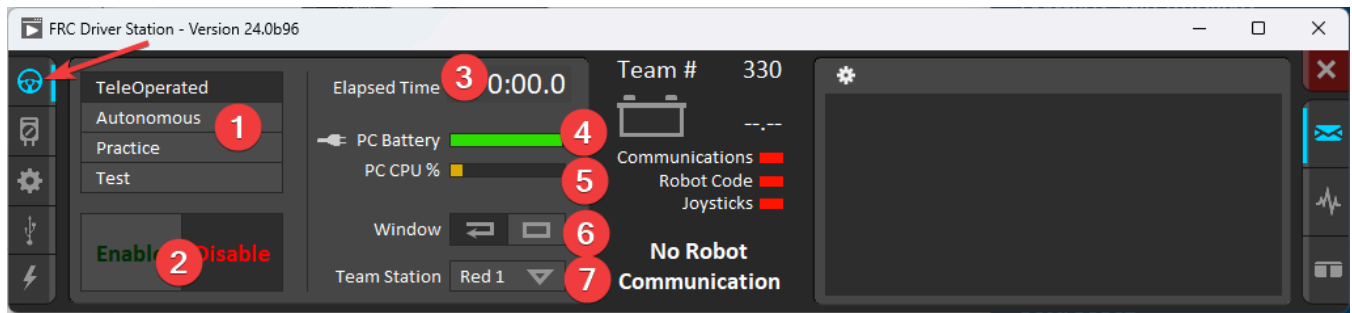
3. Major Status Indicators - These three indicators display major status items for the DS. The "Communications" indicates whether the DS is currently communicating with the FRC Network Communications Task on the roboRIO (it is split in half for the TCP and UDP communication). The "Robot Code" indicator shows whether the team Robot Code is currently running (determined by whether or not the Driver Station Task in the robot code is updating the battery voltage), The "Joysticks" indicator shows if at least one joystick is plugged in and recognized by the DS.

Major Status Indicators——这三个指示器显示 DS 的主要状态项。“Communications”表示 DS 当前是否正在与 roboRIO 上的 FRC Network Communications Task 通信；该指示器分为两半，分别对应 TCP 和 UDP 通信。“Robot Code”指示器显示队伍 Robot Code 当前是否正在运行，其判断依据是 Robot Code 中的 Driver Station Task 是否正在更新电池电压。“Joysticks”指示器显示是否至少有一个 Joystick 已插入并被 DS 识别。

4. Status String - The Status String provides an overall status message indicating the state of the robot. Some examples are "No Robot Communication", "No Robot Code", "Emergency Stopped", and "Teleoperated Enabled". When the roboRIO brownout is triggered this will display "Voltage Brownout".

Status String——Status String 提供一条表示机器人状态的总体状态消息，例如“No Robot Communication”“No Robot Code”“Emergency Stopped”和“Teleoperated Enabled”。触发 roboRIO 欠压时，此处会显示“Voltage Brownout”。

Operation Tab / Operation Tab



The Operations Tab is used to control the mode of the robot and provide additional key status indicators while the robot is running.

Operations Tab 用于控制机器人模式，并在机器人运行时提供其他关键状态指示器。

1. Robot Mode - This section controls the Robot Mode.

Robot Mode——此部分用于控制 Robot Mode。

- Teleoperated Mode causes the robot to run the code in the Teleoperated portion of the match.
Teleoperated Mode 使机器人运行比赛 Teleoperated 阶段的代码。
- Autonomous Mode causes the robot to run the code in the Autonomous portion of the match.
Autonomous Mode 使机器人运行比赛 Autonomous 阶段的代码。
- Practice Mode causes the robot to cycle through the same transitions as an FRC match after the Enable button is pressed (timing for practice mode can be found on the setup tab). When Practice Mode is in use, the DS will flash the background orange to indicate a pending enable (either the start of Autonomous or the start of Teleop after an A-Stop).

按下 Enable Button 后，Practice Mode 会使机器人依次经历与 FRC 比赛相同的模式转换；Practice Mode 的时序设置位于 Setup Tab。使用 Practice Mode 时，DS 背景会闪烁橙色，表示即将启用机器人，包括 Autonomous 开始时，或 A-Stop 后 Teleop 开始时。

- Test Mode is an additional mode where test code that doesn't run in a regular match can be tested.

Test Mode 是一种附加模式，可用于测试在常规比赛中不会运行的测试代码。

2. Enable/Disable - These controls enable and disable the robot. See also Driver Station Key Shortcuts.

Enable/Disable——这些控件用于启用和禁用机器人。另请参阅“Driver Station Key Shortcuts”。

3. Elapsed Time - Indicates the amount of time the robot has been enabled.

Elapsed Time——表示机器人已启用的时长。

4. PC Battery - Indicates current state of DS PC battery and whether the PC is plugged in.

PC Battery——表示 DS PC 电池的当前状态，以及 PC 是否已接通电源。

5. PC CPU% - Indicates the CPU Utilization of the DS PC.

PC CPU%——表示 DS PC 的 CPU 使用率。

6. Window Mode - When not on the Driver account on the Classmate allows the user to toggle between floating (arrow) and docked (rectangle).

Window Mode——未在 Classmate 上使用 Driver Account 时，允许用户在浮动模式（箭头）和停靠模式（矩形）之间切换。

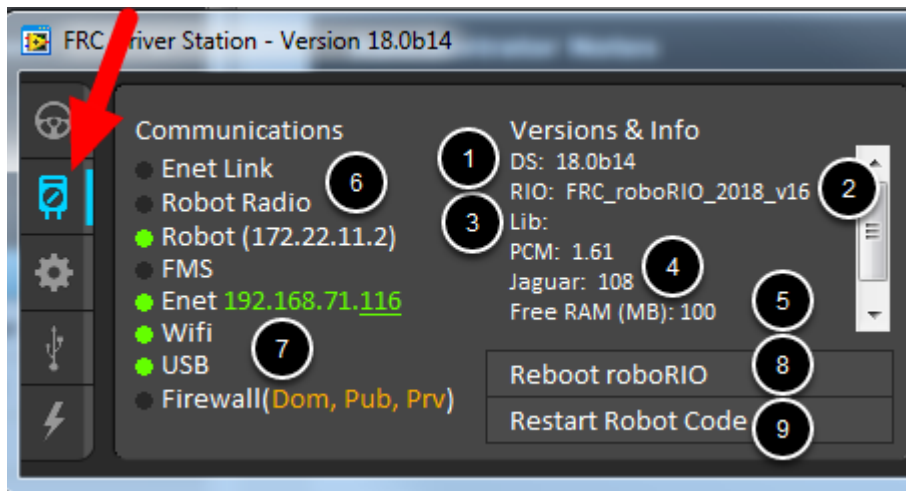
7. Team Station - When not connected to FMS, sets the team station to transmit to the robot.

Team Station——未连接到 FMS 时，设置要向机器人发送的 Team Station。

NOTE: When connected to the Field Management System the controls in sections 1 and 2 will be replaced by the words FMS Connected and the control in Section 7 will be greyed out.

说明：连接到 Field Management System 时，第 1 和第 2 部分中的控件将替换为“FMS Connected”文字，第 7 部分中的控件将变为灰色并不可用。

Diagnostics Tab / Diagnostics Tab



The Diagnostics Tab contains additional status indicators that teams can use to diagnose issues with their robot:

Diagnostics Tab 包含其他状态指示器，队伍可以利用它们诊断机器人问题：

1. DS Version - Indicates the Driver Station Version number.

DS Version——表示 Driver Station 版本号。

2. roboRIO Image Version - String indicating the version of the roboRIO Image.

roboRIO Image Version——表示 roboRIO Image 版本的字符串。

3. WPILib Version - String indicating the version of WPILib in use.

WPILib Version——表示当前所用 WPILib 版本的字符串。

4. CAN Device Versions - String indicating the firmware version of devices connected to the CAN bus. These items may not be present if the CTRE Phoenix Framework has not been loaded.

CAN Device Versions——表示连接到 CAN bus 的设备固件版本的字符串。如果未加载 CTRE Phoenix Framework，这些项目可能不会出现。

5. Memory Stats - This section shows stats about the roboRIO memory.

Memory Stats——此部分显示有关 roboRIO 内存的统计信息。

6. Connection Indicators - The top half of these indicators show connection status to various components.

Connection Indicators——这些指示器的上半部分显示与不同组件的连接状态。

- o "Enet Link" indicates the computer has something connected to the ethernet port.

"Enet Link"表示计算机的 Ethernet Port 已连接设备。

- o "Robot Radio" indicates the ping status to the robot wireless bridge at 10.XX.YY.1.

“Robot Radio”表示对位于 10.XX.YY.1 的机器人 Wireless Bridge 的 Ping 状态。

- o "Robot" indicates the ping status to the roboRIO using mDNS (with a fallback of a static 10.TE.AM.2 address).

“Robot”表示使用 mDNS 对 roboRIO 的 Ping 状态；如果 mDNS 不可用，则回退到静态地址 10.TE.AM.2。

- o "FMS" indicates if the DS is receiving packets from FMS (this is NOT a ping indicator).

“FMS”表示 DS 是否正在接收来自 FMS 的 Packet；这不是 Ping 指示器。

7. Network Indicators - The second section of indicators indicates status of network adapters and firewalls. These are provided for informational purposes; communication may be established even with one or more unlit indicators in this section.

Network Indicators——第二部分指示器显示 Network Adapter 和 Firewall 的状态。这些指示器仅供参考；即使本部分中有一个或多个指示器未点亮，也可能已建立通信。

- o "Enet" indicates the IP address of the detected Ethernet adapter

“Enet”表示检测到的 Ethernet Adapter 的 IP Address。

- o "WiFi" indicates if a wireless adapter has been detected as enabled

“WiFi”表示是否检测到已启用的 Wireless Adapter。

- o "USB" indicates if a roboRIO USB connection has been detected

“USB”表示是否检测到 roboRIO USB 连接。

- o "Firewall" indicates if any firewalls are detected as enabled. Enabled firewalls will show in orange (Dom = Domain, Pub = Public, Prv = Private)

“Firewall”表示是否检测到任何已启用的 Firewall。已启用的 Firewall 会以橙色显示（Dom = Domain、Pub = Public、Prv = Private）。

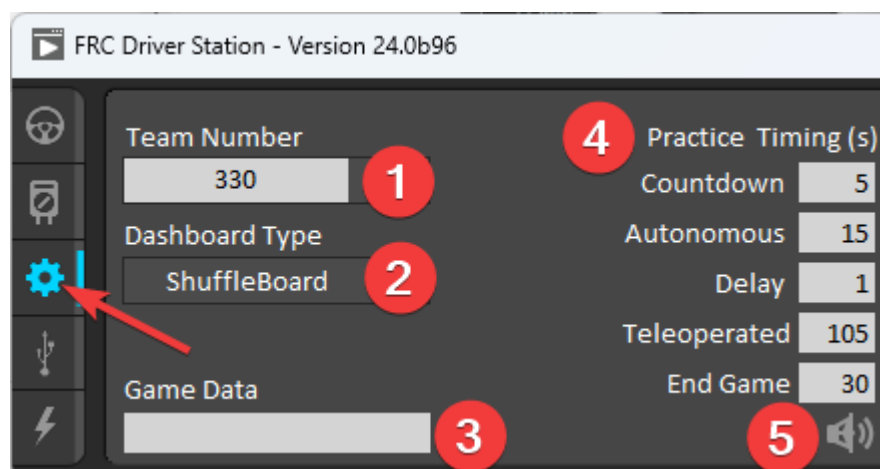
8. Reboot roboRIO - This button attempts to perform a remote reboot of the roboRIO (after clicking through a confirmation dialog).

Reboot roboRIO ——单击确认对话框后，此按钮会尝试远程重新启动 roboRIO。

9. Restart Robot Code - This button attempts to restart the code running on the robot (but not restart the OS).

Restart Robot Code ——此按钮会尝试重新启动机器人上运行的代码，但不会重新启动 OS。

Setup Tab / Setup Tab



The Setup Tab contains a number of buttons teams can use to control the operation of the Driver Station:

Setup Tab 包含多个按钮，队伍可以用它们控制 Driver Station 的运行：

1. **Team Number** - Should contain your FRC Team Number. This controls the mDNS name that the DS expects the robot to be at. Shift clicking on the dropdown arrow will show all roboRIO names detected on the network for troubleshooting purposes.

Team Number ——应填写 FRC 队伍编号。此设置控制 DS 预期机器人所使用的 mDNS Name。按住 Shift 并单击下拉箭头，将显示网络中检测到的所有 roboRIO Name，以便进行故障排查。

2. **Dashboard Type** - Controls what Dashboard is launched by the Driver Station. **Default** launches the file pointed to by the "FRC DS Data Storage.ini" (for more information about setting a [custom dashboard](#)). By default this is Dashboard.exe in the Program Files (x86)\FRC Dashboard folder.

SmartDashboard and **Shuffleboard** launch the respective dashboards included with the Java WPILib installation.

Dashboard Type ——控制 Driver Station 启动哪个 Dashboard。**Default** 启动“FRC DS Data Storage.ini”所指向的文件；有关设置[自定义 Dashboard](#)的更多信息，请参阅相应文档。默认情况下，该文件是 Program Files (x86)\FRC Dashboard 文件夹中的 Dashboard.exe。**SmartDashboard** 和 **Shuffleboard** 分别启动 Java WPILib 安装中包含的对应 Dashboard。

3. **Game Data** - This box can be used for at home testing of the Game Data API. Text entered into this box will appear in the Game Data API on the Robot Side. When connected to FMS, this data will be populated by the field automatically.

Game Data ——此输入框可用于在家测试 Game Data API。输入其中的文字会出现在 Robot Side 的 Game Data API 中。连接到 FMS 时，此数据将由场地自动填充。

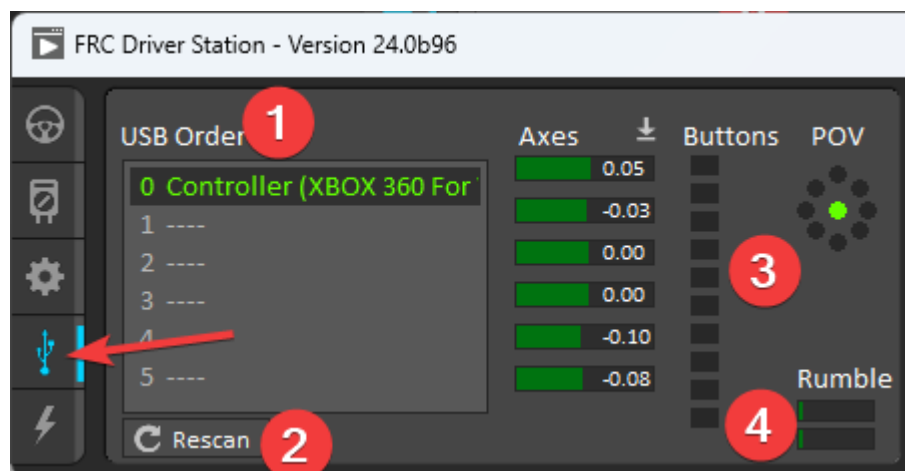
4. Practice Mode Timing - These boxes control the timing of each portion of the practice mode sequence. When the robot is enabled in practice mode the DS automatically proceeds through the modes indicated from top to bottom.

Practice Mode Timing——这些输入框控制 Practice Mode 序列中各阶段的时长。在 Practice Mode 下启用机器人后，DS 会按照从上到下的顺序自动切换到所示模式。

5. Audio Control - This button controls whether audio tones are sounded when the Practice Mode is used.

Audio Control——此按钮控制使用 Practice Mode 时是否播放提示音。

USB Devices Tab / USB Devices Tab



The USB Devices tab includes the information about the USB Devices connected to the DS

USB Devices Tab 包含连接到 DS 的 USB Device 信息。

1. USB Setup List - This contains a list of all compatible USB devices connected to the DS. Pressing a button on a device will highlight the name in green and put 2 *s before the device name

USB Setup List——其中列出了连接到 DS 的所有兼容 USB Device。按下设备上的某个按钮会以绿色突出显示设备名称，并在设备名称前添加 2 个 *。

2. **Rescan** - This button will force a Rescan of the USB devices. While the robot is disabled, the DS will automatically scan for new devices and add them to the list. To force a complete re-scan or to re-scan while the robot is Enabled (such as when connected to FMS during a match) press F1 or use this button.

Rescan——此按钮会强制重新扫描 USB Device。机器人处于禁用状态时，DS 会自动扫描新设备并将其添加到列表中。若要强制执行完整的重新扫描，或在机器人处于启用状态时重新扫描（例如比赛期间连接到 FMS 时），请按 F1 或使用此按钮。

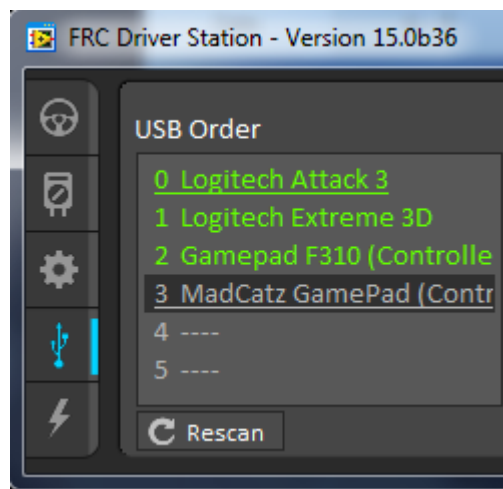
3. Device indicators - These indicators show the current status of the Axes, buttons, and POV of the joystick.

Device Indicators——这些指示器显示 Joystick 的 Axis、Button 和 POV 的当前状态。

4. Rumble - For XInput devices (such as X-Box controllers) the Rumble control will appear. This can be used to test the rumble functionality of the device. The top bar is "Right Rumble" and the bottom bar is "Left Rumble". Clicking and holding anywhere along the bar will activate the rumble proportionally (left is no rumble = 0, right is full rumble = 1). This is a control only and will not indicate the Rumble value set in robot code.

Rumble——对于 XInput Device（例如 X-Box Controller），会显示 Rumble Control，可用于测试设备的振动功能。上方条形控件是“Right Rumble”，下方条形控件是“Left Rumble”。在条形控件任意位置单击并按住，会按比例启用振动（最左侧表示无振动 = 0，最右侧表示完全振动 = 1）。该控件仅用于控制，不会显示 Robot Code 中设置的 Rumble Value。

Re-Arranging and Locking Devices / 重新排列和锁定设备



The Driver Station has the capability of "locking" a USB device into a specific slot. This is done automatically if the device is dragged to a new position and can also be triggered by double clicking on the device. "Locked" devices will show up with an underline under the device. A locked device will reserve its slot even when the device is not connected to the computer (shown as grayed out and underlined). Devices can be unlocked (and unconnected devices removed) by double clicking on the entry.

Driver Station 可以将 USB Device“锁定”到特定 Slot。将设备拖动到新位置时会自动执行锁定，也可以双击设备触发锁定。“Locked”设备的名称下方会显示下划线。即使锁定的设备未连接到计算机，其 Slot 仍会被保留；此时设备显示为灰色并带下划线。双击条目可以解锁设备，也可以移除未连接的设备。

NOTE: If you have two or more of the same device, they should maintain their position as long as all devices remain plugged into the computer in the same ports they were locked in. If you switch the ports of two identical devices the lock should follow the port, not the device. If you re-arrange the ports (take one device and plug it into a new port instead of swapping) the behavior is not determinate (the devices may swap slots). If you unplug one or more of the set of devices, the positions of the others may move; they should return to the proper locked slots when all devices are reconnected.

说明：如果有两个或更多相同设备，只要所有设备始终连接到锁定时所用的同一计算机 Port，它们就应保持各自位置。如果交换两个相同设备所连接的 Port，锁定状态应跟随 Port，而不是设备。如果重新排列 Port，即将某个设备插入新的 Port 而不是交换两个设备，其行为并不确定，设备可能互换 Slot。如果拔出这组设备中的一个或多个，其他设备的位置可能发生变化；重新连接所有设备后，它们应返回正确的锁定 Slot。

Example: The image above shows 4 devices:

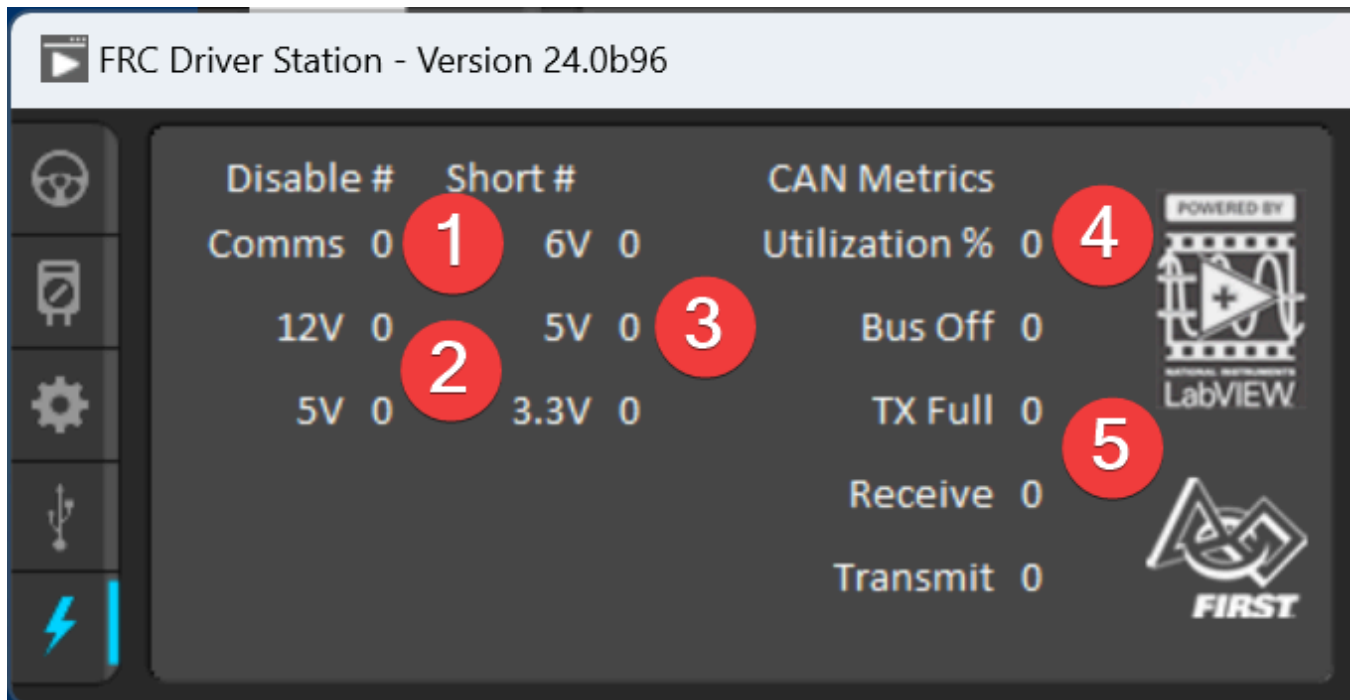
示例：上图显示了 4 个设备：

- A Locked "Logitech Attack 3" joystick. This device will stay in this position unless dragged somewhere else or unlocked
已锁定的“Logitech Attack 3”Joystick。除非将其拖动到其他位置或解除锁定，否则该设备会保持在此位置。
- An unlocked "Logitech Extreme 3D" joystick
未锁定的“Logitech Extreme 3D”Joystick。
- An unlocked "Gamepad F310 (Controller)" which is a Logitech F310 gamepad
未锁定的“Gamepad F310 (Controller)”，即 Logitech F310 Gamepad。
- A Locked, but disconnected "MadCatz GamePad (Controller)" which is a MadCatz Xbox 360 Controller
已锁定但未连接的“MadCatz GamePad (Controller)”，即 MadCatz Xbox 360 Controller。

In this example, unplugging the Logitech Extreme 3D joystick will result in the F310 Gamepad moving up to slot 1. Plugging in the MadCatz Gamepad (even if the devices in Slots 1 and 2 are removed and those slots are empty) will result in it occupying Slot 3.

在此示例中，拔出 Logitech Extreme 3D Joystick 会使 F310 Gamepad 上移到 Slot 1。插入 MadCatz Gamepad 后，即使 Slot 1 和 2 中的设备已移除且这些 Slot 为空，它仍会占用 Slot 3。

CAN/Power Tab / CAN/Power Tab



The last tab on the left side of the DS is the CAN/Robot Power Tab. This tab contains information about the power status of the roboRIO and the status of the CAN bus:

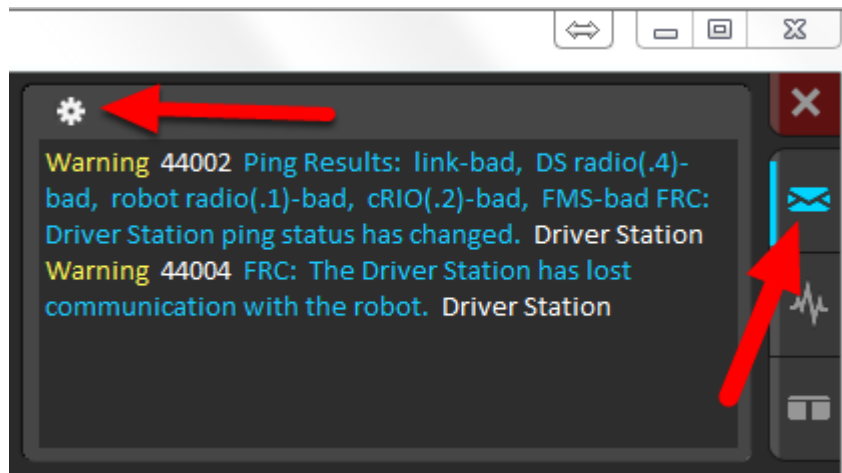
DS 左侧最后一个 Tab 是 CAN/Robot Power Tab。该 Tab 包含 roboRIO 电源状态和 CAN bus 状态的信息：

1. Comms Faults - Indicates the number of Comms faults that have occurred since the DS has been connected
Comms Faults——表示 DS 连接以来发生的 Comms Fault 数量。
2. 12V Faults - Indicates the number of input power faults (Brownouts) that have occurred since the DS has been connected
12V Faults——表示 DS 连接以来发生的输入电源故障（欠压）数量。
3. 6V/5V/3.3V Faults - Indicates the number of faults (typically caused by short circuits) that have occurred on the User Voltage Rails since the DS has been connected
6V/5V/3.3V Faults——表示 DS 连接以来 User Voltage Rail 上发生的故障数量，这些故障通常由短路造成。
4. CAN Bus Utilization - Indicates the percentage utilization of the CAN bus
CAN Bus Utilization——表示 CAN bus 的使用率百分比。
5. CAN faults - Indicates the counts of each of the 4 types of CAN faults since the DS has been connected
CAN Faults——表示 DS 连接以来 4 种 CAN Fault 各自的发生次数。

If a fault is detected, the indicator for this tab (shown in blue in the image above) will turn red.

如果检测到故障，此 Tab 的指示器（在上图中显示为蓝色）将变为红色。

Messages Tab / Messages Tab



The Messages tab displays diagnostic messages from the DS, WPILib, User Code, and/or the roboRIO.

Messages Tab 显示来自 DS、WPILib、User Code 和/或 roboRIO 的诊断消息。

To access settings for the Messages tab, click the Gear icon. This will display a menu that will allow you to clear the box, launch a larger Console window for viewing messages, launch the DS Log Viewer, launch a viewer for program timing, or download log files from the robot.

若要访问 Messages Tab 的设置，请单击 Gear Icon。随后会显示一个菜单，可用于清空消息框、打开用于查看消息的较大 Console Window、启动 DS Log Viewer、启动 Program Timing Viewer，或从机器人下载 Log File。

Charts Tab / Charts Tab



The Charts tab plots and displays advanced indicators of robot status to help teams diagnose robot issues:

Charts Tab 绘制并显示机器人状态的高级指示器，帮助队伍诊断机器人问题：

1. The top graph charts trip time in milliseconds in green (against the axis on the right) and lost packets per second in orange (against the axis on the left).

上方图表以绿色绘制以毫秒为单位的 Trip Time（对应右侧坐标轴），并以橙色绘制每秒丢失的 Packet 数量（对应左侧坐标轴）。

2. The bottom graph plots battery voltage in yellow (against the axis on the left), roboRIO CPU in red (against the axis on the right), DS Requested mode as a continuous line on the bottom of the chart and robot mode as a discontinuous line above it.

下方图表以黄色绘制电池电压（对应左侧坐标轴），以红色绘制 roboRIO CPU（对应右侧坐标轴），在图表底部以连续线绘制 DS Requested Mode，并在其上方以不连续线绘制 Robot Mode。

3. This key shows the colors used for the DS Requested and Robot Reported modes in the bottom chart.

此图例显示下方图表中 DS Requested Mode 和 Robot Reported Mode 所使用的颜色。

4. Chart scale - These controls change the time scale of the DS Charts.

Chart Scale——这些控件用于更改 DS Chart 的时间尺度。

5. This button launches the [DS Log File Viewer](#).

此按钮启动 [DS Log File Viewer](#)。

The DS Requested mode is the mode that the Driver Station is commanding the robot to be in. The Robot Reported mode is what code is actually running based on reporting methods contained in the coding frameworks for each language.

DS Requested Mode 是 Driver Station 命令机器人进入的模式。Robot Reported Mode 则根据各语言编程框架中包含的报告机制，表示实际正在运行的代码模式。

Both Tab / Both Tab

The last tab on the right side is the Both tab which displays Messages and Charts side by side.

右侧最后一个 Tab 是 Both Tab，它并排显示 Messages 和 Charts。

Driver Station (Operator Console) Best Practices / Driver Station (Operator Console) 最佳实践

Want to ensure the driver station isn't a stopper for your team at the FIRST Robotics Competition (FRC®) field? Building and configuring a solid driver station laptop and peripherals is an easy project. Read on to find lessons learned by many teams over thousands of matches.

想确保 Driver Station 不会成为队伍在 FIRST Robotics Competition (FRC®) 场地上的障碍吗？构建并配置一套可靠的 Driver Station Laptop 和外围设备并不困难。请继续阅读，了解许多队伍经过数千场比赛总结出的经验。

Selecting hardware / 选择硬件

Dedicate a laptop to be used solely for driving. Many teams do. A dedicated machine allows you manage the configuration for one goal – being ready to compete at the field. Dedicated means no other software except the FRC Driver Station application and associated Dashboard installed or running.

准备一台专门用于操控机器人的 Laptop，许多队伍都会这样做。专用设备使你能够围绕一个目标管理其配置——随时准备在赛场上比赛。“专用”意味着除了 FRC Driver Station 应用程序及相关 Dashboard 外，不安装或运行其他软件。

Use a business-class laptop for your driver station laptop. Why? They're much more durable than the \$300 Black Friday special at Best Buy. They'll survive being banged around at the competition. Business-class laptops have higher quality device drivers, and the drivers are maintained for a longer period than consumer laptops. This makes your investment last longer. Lenovo ThinkPad T series and Dell Latitude are two popular business-class brands you'll commonly see at competitions.

Driver Station Laptop 应使用商用级 Laptop。原因是它们比 Best Buy 黑色星期五特价 \$300 的产品耐用得多，能够承受比赛中的磕碰。商用级 Laptop 的 Device Driver 质量更高，而且其维护周期比消费级 Laptop 更长，因此投资能够使用更久。Lenovo ThinkPad T 系列和 Dell Latitude 是比赛中常见的两种商用级产品。

Consider used laptops rather than new. The FRC Driver Station application and dashboard software uses very few system resources, so you don't need to buy a new laptop -- instead, buy a cheap 4-5 year old used one. There are thousands for sale every day on eBay. You might even get one donated by a used computer store.

可以考虑购买二手 Laptop，而不是新设备。FRC Driver Station 应用程序和 Dashboard 软件占用的系统资源很少，因此无需购买新 Laptop；可以买一台价格低廉、使用了 4-5 年的二手设备。eBay 每天都有数千台此类设备出售，甚至还可能从二手电脑商店获得捐赠。

Laptop recommended features / Laptop 推荐特性

1. 8GB of RAM or greater

8GB 或更大的 RAM。

2. A display size of 14" or greater, with 1080p minimum resolution

屏幕尺寸不小于 14"，最低分辨率为 1080p。

3. Ports

Port:

- a. A built-in Ethernet port is highly preferred. Ensure that it's a full-sized port. The hinged Ethernet ports

don't hold up to repeated use.

强烈建议使用内置 Ethernet Port。确认它是全尺寸 Port；铰链式 Ethernet Port 经不起反复使用。

b. If the Ethernet port on your laptop is dodgy, either replace the laptop (recommended) or buy a USB Ethernet dongle from a reputable brand. See Recommended USB Ethernet adapters for a list. Many teams find that USB Ethernet is less reliable than built-in Ethernet, primarily due to cheap hardware and bad drivers. The dongles given to rookies in the KOP have a reputation for working well.

如果 Laptop 的 Ethernet Port 不可靠，可以更换 Laptop（推荐），或购买信誉良好品牌的 USB Ethernet Dongle。相关列表见“Recommended USB Ethernet adapters”。许多队伍发现 USB Ethernet 的可靠性低于内置 Ethernet，主要原因是廉价硬件和质量较差的 Driver。KOP 中提供给 Rookie 的 Dongle 以工作可靠而闻名。

c. 2 USB ports minimum. These can be USB-A or USB-C ports, depending on the peripherals you're using.

至少需要 2 个 USB Port。根据所使用的外围设备，它们可以是 USB-A 或 USB-C Port。

4. A keyboard. It's hard to quickly do troubleshooting on touch-only computers at the field.

键盘。在赛场上很难使用仅支持触控的计算机快速进行故障排查。

5. A solid-state disk (SSD), 256 GB or larger. If the laptop has a rotating disk, spend \$50 and replace it with an SSD.

256 GB 或更大的 Solid-State Disk (SSD)。如果 Laptop 使用机械硬盘，可以花费 \$50 将其更换为 SSD。

6. Supports Windows 11. Given that it is at end of life, don't plan on support for Windows 10 beyond the 2026 competition year.

支持 Windows 11。Windows 10 已进入生命期末期，因此不要指望在 2026 比赛年度之后继续获得对 Windows 10 的支持。

7. AMD or Intel CPU. The FRC Driver Station Application does not work on machines with ARM CPUs (e.g., Snapdragon).

AMD 或 Intel CPU。FRC Driver Station Application 无法在采用 ARM CPU（例如 Snapdragon）的计算机上运行。

8. A 2.4 and 5 GHz Wi-Fi radio. 6GHz support is optional but not necessary. For the 2026 game, the [preferred practice configuration](#) does not require 6GHz, and the 2027 SystemCore has built-in 5 GHz connectivity.

支持 2.4 和 5 GHz 的 Wi-Fi Radio。6GHz 支持是可选项，并非必需。对于 2026 赛季，[推荐的练习配置](#)不要求 6GHz，而 2027 SystemCore 内置 5 GHz 连接能力。

Building your driver station / 构建 Driver Station



Two level driver station with power and USB hub in lower level. USB-C ethernet adapter provides field connectivity.

双层 Driver Station，下层安装电源和 USB Hub。USB-C Ethernet Adapter 提供场地连接。



Driver station with integrated controller holders. Controllers and field network are directly connected to laptop.

集成 Controller Holder 的 Driver Station。Controller 和场地网络直接连接到 Laptop。



Single level driver station with external button controls for task automation. USB-C ethernet/hub combo provides field connectivity.

单层 Driver Station，带有用于任务自动化的外部按钮控件。USB-C Ethernet/Hub 组合设备提供场地连接。

It's best if you combine your driver station laptop and all needed peripherals into a [single unit](#).

最好将 Driver Station Laptop 与所有必要的外围设备组合成一个[整体](#)。

Use an Ethernet port saver to make your Ethernet connection. This extends the life of the port on the laptop. This is particularly important if you have a consumer-grade laptop with a hinged Ethernet port.

使用 Ethernet Port Saver 建立 Ethernet 连接，这可以延长 Laptop 上 Port 的使用寿命。对于带铰链式 Ethernet Port 的消费级 Laptop，这一点尤其重要。

Robots often hit the driver station wall with large forces, which can push your driver station off the shelf. Put hook tape on the bottom of the driver station to reduce the chance of that happening. The Game Manual has details on the correct location of the tape. Shorter driver stations have lower center of gravity and are more likely to stay on the shelf.

机器人经常以很大的力撞击 Driver Station Wall，可能会把 Driver Station 从架子上推落。在 Driver Station 底部粘贴 Hook Tape，可以降低这种情况发生的概率。Game Manual 中详细说明了胶带的正确位置。高度较低的 Driver Station 重心更低，更容易保持在架子上。

When fastening down peripherals, it's best to use hook and loop tape. This allows quick changes at the field when needed.

固定外围设备时，最好使用 Hook-and-Loop Tape，以便在赛场上需要时快速调整。

Ensure that the USB ports on the laptop are accessible.

确保 Laptop 上的 USB Port 便于使用。

Setting up the laptop / 设置 Laptop

Once you've acquired your laptop, here's a list of things to do.

获得 Laptop 后，请完成以下操作。

1. If you're buying used, ensure you're starting from a fresh Windows install.

如果购买二手设备，请确保从全新安装的 Windows 开始使用。

2. Create one account to use both for software installation and driving. This account must be a member of the Administrator group.

创建一个同时用于安装软件和操控机器人的 Account。该 Account 必须属于 Administrator Group。

3. Remove any 3rd party antivirus or antimalware software. Instead, use Windows Defender. Since you're only connecting to the internet for Windows and FRC software updating, the risk is low.

删除所有第三方 Antivirus 或 Antimalware 软件，改用 Windows Defender。由于仅在更新 Windows 和 FRC 软件时连接 Internet，因此风险较低。

4. Only install software on your driver station laptop that's needed for driving. Your goal here is to eliminate variables that might interfere with proper operation. Don't use the driver station laptop for programming. Remove any unneeded preinstalled software ("bloatware") that came with the machine. Don't use the laptop for gaming back at the hotel the night before the event.

Driver Station Laptop 上只安装操控机器人所需的软件，目标是消除可能干扰正常运行的变量。不要使用 Driver Station Laptop 编程。删除设备附带的所有非必要预装软件（“bloatware”）。比赛前一晚回到酒店后，也不要使用这台 Laptop 玩游戏。

5. Avoid managed Windows installations from the school's IT department. These deployments are built for the school environment and often come with unwanted software that interferes with your robot's operation.

避免使用由学校 IT 部门管理的 Windows 安装。这类部署是针对学校环境构建的，通常附带可能干扰机器人运行的非必要软件。

6. Make these power settings for both the Plugged in and On battery profiles, by navigating to Start -> Settings -> System -> Power and battery

依次进入 Start -> Settings -> System -> Power and battery，并为 Plugged in 和 On battery 两种配置文件进行以下电源设置：

- a. Set "Make my device sleep after" to Never for both Plugged in and On battery

将 Plugged in 和 On battery 的“Make my device sleep after”都设置为 Never。

- b. Turn off Energy Saver

关闭 Energy Saver。

- c. Set Power Mode to Best Performance

将 Power Mode 设置为 Best Performance。

- d. Ensure the laptop battery can hold a charge for at least an hour after making the changes above. This allows plenty of time for the robot and drive team to go through the queue and reach the alliance station without mains power.

完成上述更改后，确保 Laptop 电池至少能够维持一小时。这可以让机器人和 Drive Team 在没有市电的情况下，有充足时间完成排队并到达 Alliance Station。

7. If you are running a non-English version of Windows, set the user to English prior to doing any software installation. If your laptop has a non-English Single Language Edition, you should install an edition of Windows that is supported by the FRC Driver Station application and supports English.

如果运行的是非英文版 Windows，请在安装任何软件之前将用户语言设置为 English。如果 Laptop 使用非英文 Single Language Edition，应安装受 FRC Driver Station Application 支持且支持 English 的 Windows 版本。

Prior to leaving for the competition / 前往比赛前

Items in this section require internet connectivity and should be done prior to arriving at the event.

本节中的操作需要 Internet 连接，应在到达活动现场前完成。

1. Install all Windows updates, then test the driver station with your robot. Keep running Windows Update until no updates show as pending.

安装所有 Windows Update，然后使用机器人测试 Driver Station。持续运行 Windows Update，直到不再显示 Pending Update。

2. Pause updates. Navigate to Start -> Settings -> Windows Update, then set Pause updates to cover the time of the competition.

暂停更新。依次进入 Start -> Settings -> Windows Update，然后设置 Pause updates，使暂停时间覆盖比赛期间。

3. Change "Active Hours" for Windows Updates to prevent updates from installing during competition hours. This helps if you forget to pause updates. Navigate to Start -> Settings -> Windows Update -> Advanced options, then select Active Hours. Be sure to take timezone differences at your competition location into account.

更改 Windows Update 的“Active Hours”，防止在比赛时间安装更新。如果忘记暂停更新，此设置可以提供保护。依次进入 Start -> Settings -> Windows Update -> Advanced options，然后选择 Active Hours。务必考虑比赛地点的时区差异。

4. Bring a trusted USB and Ethernet cable for use connecting to the roboRIO.

携带可靠的 USB Cable 和 Ethernet Cable，用于连接 roboRIO。

At The Competition / 比赛现场

1. Turn off Windows firewall using the instructions in the WPILib documentation.

按照 WPILib 文档中的说明关闭 Windows Firewall。

2. Turn off the Wi-Fi adapter, either using the dedicated hardware Wi-Fi switch or by using the Wi-Fi control in Quick Settings.

使用专用硬件 Wi-Fi Switch，或 Quick Settings 中的 Wi-Fi Control，关闭 Wi-Fi Adapter。

3. Charge the driver station laptop when it's in the pit.

Driver Station Laptop 位于 Pit 时为其充电。

4. Ensure everyone on the drive team knows the password. You'd be surprised at how often drivers arrive at the field without knowing the password.

确保 Drive Team 中每个人都知道密码。Driver 到达赛场却不知道密码的情况比想象中更常见。

5. Limit web browsing to FRC related web sites. This minimizes the chance of getting malware during the competition.

将 Web 浏览限制在 FRC 相关网站，以尽量降低比赛期间感染恶意软件的可能性。

6. Don't plan on using internet access at the event to make software updates. There likely won't be any in the venue, and hotel Wi-Fi varies widely in quality. If you do need updates, contact a Control System Advisor in the pit.

不要计划在活动现场使用 Internet 更新软件。场馆很可能没有可用 Internet，而且酒店 Wi-Fi 的质量差异很大。如果确实需要更新，请联系 Pit 中的 Control System Advisor。

Before Each Match / 每场比赛前

1. Make sure the laptop is on and logged in prior to the end of the match before yours.

在前一场比赛结束前，确保 Laptop 已开机并完成登录。

2. Close programs that aren't needed during the match.

关闭比赛期间不需要的程序。

3. Bring your laptop charger to the field. Power is provided in each player station. Why? Windows is less aggressive about power management when the laptop is plugged in, which means the FRC Driver Station application behaves more predictably.

将 Laptop Charger 带到赛场。每个 Player Station 都提供电源。原因是 Laptop 接通电源时，Windows 的电源管理策略没有那么激进，因此 FRC Driver Station Application 的行为更加可预测。

4. Fasten your laptop with hook-and-loop tape to the player station shelf. You never know when your alliance partner will have an autonomous programming issue and blast the wall.

使用 Hook-and-Loop Tape 将 Laptop 固定到 Player Station Shelf 上。你永远不知道 Alliance Partner 何时会因 Autonomous 编程问题而猛烈撞击墙壁。

5. Ensure joysticks and controllers are functioning.

确认 Joystick 和 Controller 工作正常。

- a. In the USB tab in the FRC Driver Station application, drag and drop to assign joysticks as needed.

在 FRC Driver Station Application 的 USB Tab 中，根据需要拖放分配 Joystick。

- b. Verify that buttons are working and joysticks return to zero position.

确认 Button 工作正常，并且 Joystick 能够回到零位。

- c. Use the rescan button (F1) if joysticks / controllers do not appear green.

如果 Joystick / Controller 未显示为绿色，请使用 Rescan Button (F1)。

- d. Use the rescan button (F1) during competition if joystick or controllers become unplugged and then are plugged back in or otherwise turn gray during competition.

比赛期间，如果 Joystick 或 Controller 被拔出后重新插入，或因其他原因变为灰色，请使用 Rescan Button (F1)。

6. Ensure your Dashboard is connected to the robot after your driver station laptop connects to the robot.

Driver Station Laptop 连接到机器人后，确认 Dashboard 也已连接到机器人。

7. Ensure FRC Driver Station application is fully visible and in focus during the match to avoid Windows deprioritizing it, and to ease troubleshooting.

比赛期间确保 FRC Driver Station Application 完全可见并具有焦点，以避免 Windows 降低其优先级，同时也

便于故障排查。

Recommended USB Ethernet adapters / 推荐的 USB Ethernet Adapter

Members of the FRC community report good results with these USB Ethernet adapters.

FRC 社区成员反馈，以下 USB Ethernet Adapter 的使用效果良好。

- Henrety USB C to Ethernet Adapter
- Siig SuperSpeed USB 3.0 LAN Hub
- TP-Link UE330
- uni USB to Ethernet Adapter with hub (both USB-A and USB-C styles)

Don't be confused and use a USB RJ-45 extender. These are reported to fry USB ports.

请勿误用 USB RJ-45 Extender。据报告，这类设备会烧坏 USB Port。



Credits / 致谢

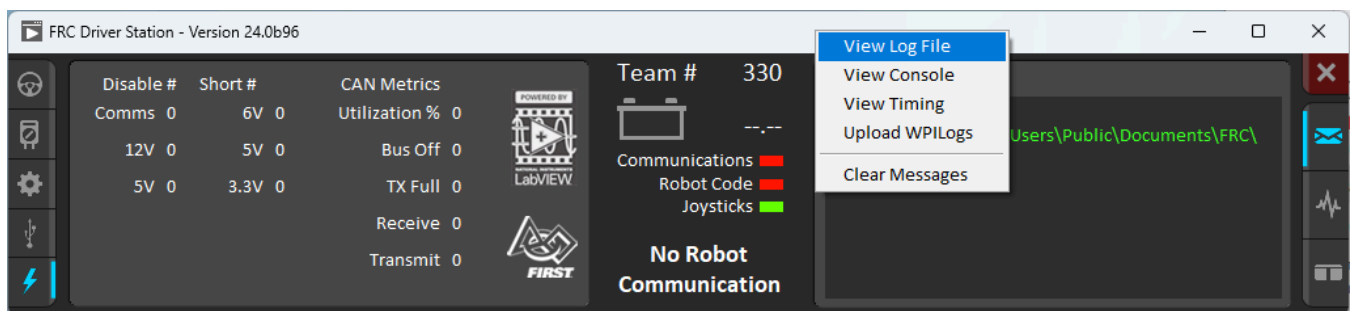
The source document for this page was created by Steve Peterson, with contributions from Juan Chong, James Cole-Henry, Rick Kosbab, Greg McKaskle, Chris Picone, Chris Roadfeldt, Joe Ross, and Ryan Sjostrand. The original post and follow-up posts can be found [here](#).

本页的源文档由 Steve Peterson 创建，Juan Chong、James Cole-Henry、Rick Kosbab、Greg McKaskle、Chris Picone、Chris Roadfeldt、Joe Ross 和 Ryan Sjostrand 参与贡献。原始帖子和后续帖子可在[此处](#)查看。

Driver Station Log File Viewer / Driver Station Log File Viewer

In an effort to provide information to aid in debugging, the FRC® Driver Station creates log files of important diagnostic data while running. These logs can be reviewed later using the FRC Driver Station Log Viewer. The Log Viewer can be found via the shortcut installed in the Start menu, in the FRC Driver Station folder in Program Files, or via the Gear icon in the Driver Station.

为了提供有助于调试的信息，FRC® Driver Station 会在运行时创建包含重要诊断数据的 Log File。之后可以使用 FRC Driver Station Log Viewer 查看这些 Log。可以通过 Start Menu 中安装的快捷方式、Program Files 中的 FRC Driver Station 文件夹，或 Driver Station 中的 Gear Icon 打开 Log Viewer。



NOTE: Several alternative tools exist that provide similar functionality to the FRC Driver Station Log Viewer. AdvantageScope is an option included in WPILib, and [DSLOG Reader](#) is a third-party option. Note that WPILib offers no support for third-party projects.

说明：还有多种替代工具能够提供与 FRC Driver Station Log Viewer 类似的功能。AdvantageScope 是 WPILib 中包含的选项，[DSLOG Reader](#) 则是第三方选项。请注意，WPILib 不为第三方项目提供支持。

Event Logs / Event Log

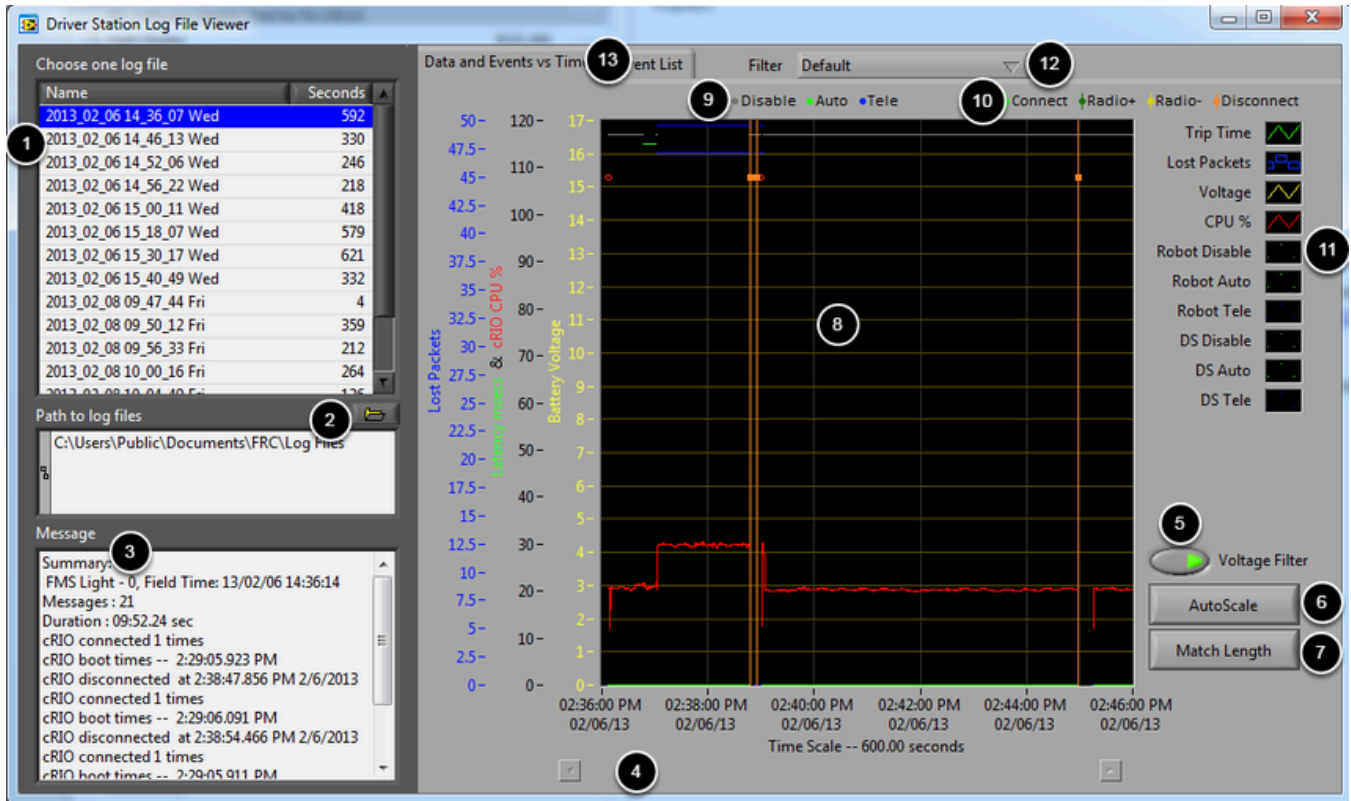
The Driver Station logs all messages sent to the Messages box on the Diagnostics tab (not the User Messages box on the Operation tab) into a new Event Log file. When viewing Log Files with the Driver Station Log File Viewer, the Event Log and Driver Station Log files are overlaid in a single display.

Driver Station 会把发送到 Diagnostics Tab 的 Messages 输入框中的所有消息记录到新的 Event Log File 中；这里所指的并不是 Operation Tab 中的 User Messages 输入框。使用 Driver Station Log File Viewer 查看 Log File 时，Event Log 和 Driver Station Log File 会叠加显示在同一界面中。

Log files are stored in `C:\Users\Public\Documents\FRC\Log Files`. Each log has date and timestamp in the file name and has two files with extension `.dslog` and `.dsevents`.

Log File 存储在 `C:\Users\Public\Documents\FRC\Log Files`。每份 Log 的文件名中都包含日期和时间戳，并由扩展名分别为 `.dslog` 和 `.dsevents` 的两个文件组成。

Log Viewer UI / Log Viewer UI



The Log Viewer contains a number of controls and displays to aid in the analysis of the Driver Station log files:

Log Viewer 包含多个控件和显示区域，用于协助分析 Driver Station Log File：

1. File Selection Box - This window displays all available log files in the currently selected folder. Click on a log file in the list to select it.

File Selection Box——此窗口显示当前所选文件夹中的所有可用 Log File。单击列表中的某个 Log File 即可将其选中。

2. Path to Log Files - This box displays the current folder the viewer is looking in for log files. This defaults to the folder that the Driver Station stores log files in. Click the folder icon to browse to a different location.

Path to Log Files——此输入框显示 Viewer 当前查找 Log File 的文件夹。默认值是 Driver Station 存储 Log File 的文件夹。单击 Folder Icon 可浏览其他位置。

3. Message Box - This box displays a summary of all messages from the Event Log. When hovering over an event on the graph this box changes to display the information for that event.

Message Box——此输入框显示 Event Log 中所有消息的摘要。将鼠标悬停在图表中的某个 Event 上时，此输入框会改为显示该 Event 的信息。

4. Scroll Bar - When the graph is zoomed in, this scroll bar allows for horizontal scrolling of the graph.

Scroll Bar——放大图表时，可以使用此 Scroll Bar 水平滚动图表。

5. Voltage Filter - This control turns the Voltage Filter on and off (defaults to on). The Voltage Filter filters out data such as CPU %, robot mode and trip time when no Battery Voltage is received (indicating that the DS is no in communication with the roboRIO).

Voltage Filter——此控件用于打开或关闭 Voltage Filter，默认处于打开状态。未收到 Battery Voltage 时，表示 DS 未与 roboRIO 通信；此时 Voltage Filter 会滤除 CPU %、Robot Mode 和 Trip Time 等数据。

6. AutoScale - This button zooms the graph out to show all data in the log.

AutoScale——此按钮缩小图表，以显示 Log 中的全部数据。

7. Match Length - This button scales the graph to approximately the length of an FRC match (2 minutes and 30 seconds shown). It does not automatically locate the start of the match, you will have to scroll using the scroll bar to locate the beginning of the Autonomous mode.

Match Length——此按钮将图表缩放到接近一场 FRC 比赛的长度，图中显示为 2 分 30 秒。它不会自动定位比赛开始位置；必须使用 Scroll Bar 滚动，找到 Autonomous Mode 的起点。

8. Graph - This display shows graph data from the DS Log file (voltage, trip time, roboRIO CPU%, Lost Packets, and robot mode) as well as overlaid event data (shown as dots on the graph with select events showing as vertical lines across the entire graph). Hovering over event markers on the graph displays information about the event in the Messages window in the bottom left of the screen.

Graph——此显示区域展示 DS Log File 中的图表数据，包括 Voltage、Trip Time、roboRIO CPU%、Lost Packet 和 Robot Mode，同时还会叠加 Event Data；Event 在图表中显示为点，部分 Event 则显示为贯穿整个图表的竖线。将鼠标悬停在图表中的 Event Marker 上，会在界面左下角的 Messages Window 中显示该 Event 的信息。

9. Robot Mode Key - Key for the Robot Mode displayed at the top of the screen

Robot Mode Key——位于界面顶部、用于说明 Robot Mode 的图例。

10. Major event key - Key for the major events, displayed as vertical lines on the graph

Major Event Key——用于说明 Major Event 的图例；Major Event 在图表中显示为竖线。

11. Graph key - Key for the graph data

Graph Key——用于说明图表数据的图例。

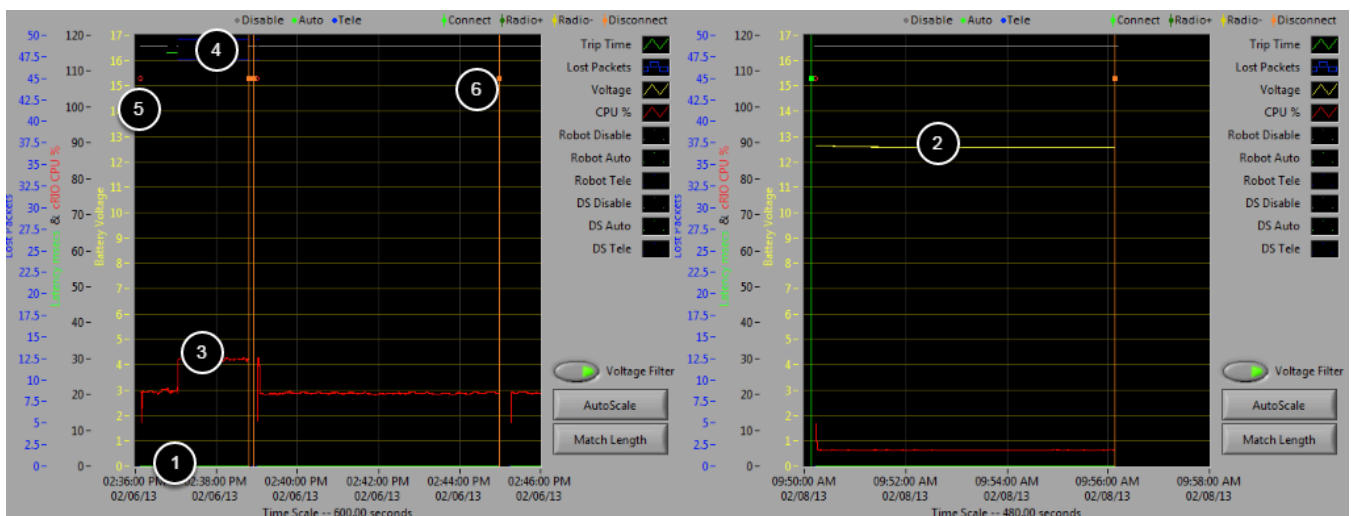
12. Filter Control - Drop-down to select the filter mode (filter modes explained below)

Filter Control——用于选择 Filter Mode 的下拉菜单；下文会说明各种 Filter Mode。

13. Tab Control - Control to switch between the Graph (Data and Events vs. Time) and Event List displays.

Tab Control——用于在 Graph（Data and Events vs. Time）与 Event List 显示之间切换的控件。

Using the Graph Display / 使用 Graph Display



The Graph Display contains the following information:

Graph Display 包含以下信息：

1. Graphs of Trip Time in ms (green line) and Lost Packets per second (displayed as blue vertical bars). In these example images Trip Time is a flat green line at the bottom of the graph and there are no lost packets

以 ms 为单位的 Trip Time 图表（绿色线）和每秒 Lost Packet 图表（蓝色竖条）。在这些示例图片中，Trip Time 是图表底部的一条水平绿色线，并且没有 Lost Packet。

2. Graph of Battery voltage displayed as a yellow line.

Battery Voltage 图表，以黄色线显示。

3. Graph of roboRIO CPU % as a red line

roboRIO CPU % 图表，以红色线显示。

4. Graph of robot mode and DS mode. The top set of the display shows the mode commanded by the Driver Station. The bottom set shows the mode reported by the robot code. In this example the robot is not reporting it's mode during the disabled and autonomous modes, but is reported during Teleop.

Robot Mode 和 DS Mode 图表。显示区域上方一组表示 Driver Station 命令的模式，下方一组表示 Robot Code 报告的模式。在此示例中，机器人在 Disabled Mode 和 Autonomous Mode 期间没有报告模式，但在 Teleop 期间报告了模式。

5. Event markers will be displayed on the graph indicating the time the event occurred. Errors will display in red; warnings will display in yellow. Hovering over an event marker will display information about the event in the Messages box at the bottom left of the screen.

图表中会显示 Event Marker，标明 Event 发生的时间。Error 显示为红色，Warning 显示为黄色。将鼠标悬停在 Event Marker 上，会在界面左下角的 Messages Box 中显示该 Event 的信息。

6. Major events are shown as vertical lines across the graph display.

Major Event 在图表显示区域中以贯穿图表的竖线表示。

To zoom in on a portion of the graph, click, and drag around the desired viewing area. You can only zoom the time axis, you cannot zoom vertically.

若要放大图表的一部分，请单击并拖动以框选所需查看区域。只能缩放时间轴，不能沿纵向缩放。

NOTE: The CTRE PDP 2.0 and AndyMark AMPD do not have built in current logging and current logs will not be available in the graph display.

说明：CTRE PDP 2.0 和 AndyMark AMPD 没有内置电流记录功能，因此 Graph Display 中不会提供 Current Log。

Event List / Event List

DS Time	Event Message Text
2:36:07.288 PM	WARNING <Code> 44007 occurred at FRC_NetworkCommunications <secondsSinceReboot> 421.365 Warning <Code> 44001 occurred at No Change to Network Configuration: "Local Area Connection"<noNIC FRC: Time since robot boot. Driver Station <time>2/6/2013 2:36:07 PM<unique#>3 ERROR <Code> -44009 occurred at Driver Station <time>2/6/2013 2:36:06 PM<unique#>2 FRC: A joystick was disconnected while the robot was enabled. Warning <Code> 44006 occurred at Driver Station <time>2/6/2013 2:36:06 PM<unique#>1 FRC: Custom I/O is not enabled or is not connected to the driver station.
2:36:07.328 PM	FMS Connected: FMS Light - 0, Field Time: 13/02/06 14:36:14
2:36:10.441 PM	WARNING <Code> 44008 occurred at FRC_NetworkCommunications <radioLostEvents> 173.563 <radioSec FRC: Robot radio detection times.
2:37:01.461 PM	Watchdog Expiration: System 1, User 0
2:38:47.856 PM	Warning <Code> 44004 occurred at Driver Station <time>2/6/2013 2:38:47 PM<unique#>4 FRC: The Driver Station has lost communication with the robot.
2:38:49.356 PM	Warning <Code> 44002 occurred at Ping Results: link-GOOD, DS radio(.4)-GOOD, robot radio(.1)-GOOD, <time>2/6/2013 2:38:49 PM<unique#>5 FRC: Driver Station ping status has changed.
2:38:53.460 PM	WARNING <Code> 44007 occurred at FRC_NetworkCommunications <secondsSinceReboot> 587.369 FRC: Time since robot boot.
2:38:54.466 PM	Warning <Code> 44004 occurred at Driver Station <time>2/6/2013 2:38:53 PM<unique#>6 FRC: The Driver Station has lost communication with the robot.
2:38:55.468 PM	Warning <Code> 44002 occurred at Ping Results: link-GOOD, DS radio(.4)-GOOD, robot radio(.1)-GOOD, <time>2/6/2013 2:38:55 PM<unique#>7 FRC: Driver Station ping status has changed.
2:38:59.278 PM	WARNING <Code> 44008 occurred at FRC_NetworkCommunications <radioLostEvents> 339.065 <radioSec FRC: Robot radio detection times. WARNING <Code> 44007 occurred at FRC_NetworkCommunications <secondsSinceReboot> 593.367

The Event List tab displays a list of events (warnings and errors) recorded by the Driver Station. The events and detail displayed are determined by the currently active filter (images shows "All Events, All Info" filter active).

Event List Tab 显示 Driver Station 记录的 Event（Warning 和 Error）列表。显示哪些 Event 及其详细程度由当前启用的 Filter 决定；图片中启用的是“All Events, All Info”Filter。

Filters / Filter

Three filters are currently available in the Log Viewer:

Log Viewer 当前提供三种 Filter：

1. Default: This filter filters out many of the errors and warnings produced by the Driver Station. This filter is useful for identifying errors thrown by the code on the Robot.

Default：此 Filter 会滤除 Driver Station 生成的许多 Error 和 Warning，有助于识别 Robot Code 抛出的 Error。
2. All Events and Time: This filter shows all events and the time they occurred

All Events and Time：此 Filter 显示所有 Event 及其发生时间。
3. All Events, All Info: This filter shows all events and all recorded info. At this time the primary difference between this filter and "All Events and Time" is that this option shows the "unique" designator for the first occurrence of a particular message.

All Events, All Info：此 Filter 显示所有 Event 和全部已记录信息。目前，此 Filter 与“All Events and Time”的主要区别在于，该选项会为某条特定消息的第一次出现显示“unique”标记。

Identifying Logs from Matches / 识别来自比赛的 Log

3:19:30.893 PM FMS Connected: Practice - 1, Field Time: 13/02/06 15:19:37

A common task when working with the Driver Station Logs is to identify which logs came from competition matches. Logs which were taken during a match can now be identified using the FMS Connected event which will display the match type (Practice, Qualification, or Elimination), match number, and the current time according to the FMS server. In this example, you can see that the FMS server time and the time of the Driver Station computer are fairly close, approximately 7 seconds apart.

处理 Driver Station Log 时，一项常见任务是识别哪些 Log 来自正式比赛。现在可以使用 FMS Connected Event 识别比赛期间记录的 Log；该 Event 会显示比赛类型（Practice、Qualification 或 Elimination）、比赛编号，以及 FMS Server 所提供的当前时间。在此示例中可以看到，FMS Server 时间与 Driver Station 计算机时间非常接近，相差约 7 秒。

Identifying Common Connection Failures with the Log Viewer / 使用 Log Viewer 识别常见连接故障

When diagnosing robot issues, there is no substitute for thorough knowledge of the system and a methodical debugging approach. If you need assistance diagnosing a connection problem at your events it is strongly recommended to seek assistance from your FTA and/or CSA. The goal of this section is to familiarize teams with how some common failures can manifest themselves in the DS Log files. Please note that depending on a variety of conditions a particular failure show slightly differently in a log file.

诊断机器人问题时，对系统的深入了解和系统化调试方法是不可替代的。如果在活动现场需要协助诊断连接问题，强烈建议向 FTA 和/或 CSA 寻求帮助。本节旨在帮助队伍熟悉一些常见故障在 DS Log File 中可能呈现的形式。请注意，受多种条件影响，同一种故障在不同 Log File 中的表现可能略有差异。

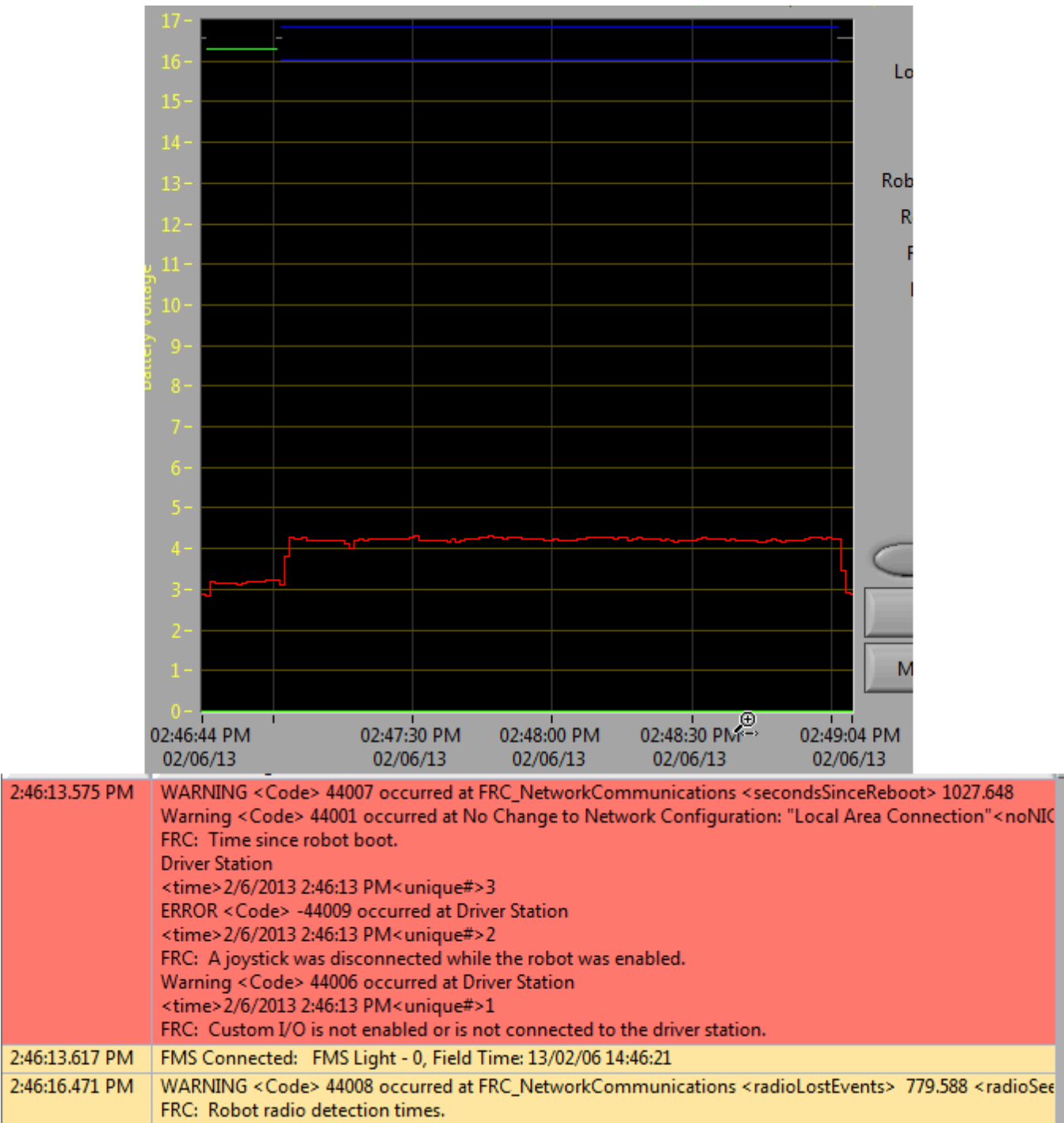
NOTE: Note that all log files shown in this section have been scaled to match length using the Match Length button and then scrolling to the beginning of the autonomous mode. Also, many of the logs do not contain battery voltage information, the platform used for log capture was not properly wired for reporting the battery voltage.

说明：本节展示的所有 Log File 都先使用 Match Length Button 缩放到比赛长度，再滚动到 Autonomous Mode 的起点。此外，许多 Log 不包含 Battery Voltage 信息，因为用于捕获 Log 的平台没有正确接线，无法报告 Battery Voltage。

TIP: Some error messages that are found in the Log Viewer are show below and more are detailed in the [driver-station-errors-warnings](#) article.

提示：下文展示了 Log Viewer 中的一些 Error Message；更多消息详见 [Driver Station Errors/Warnings](#)。

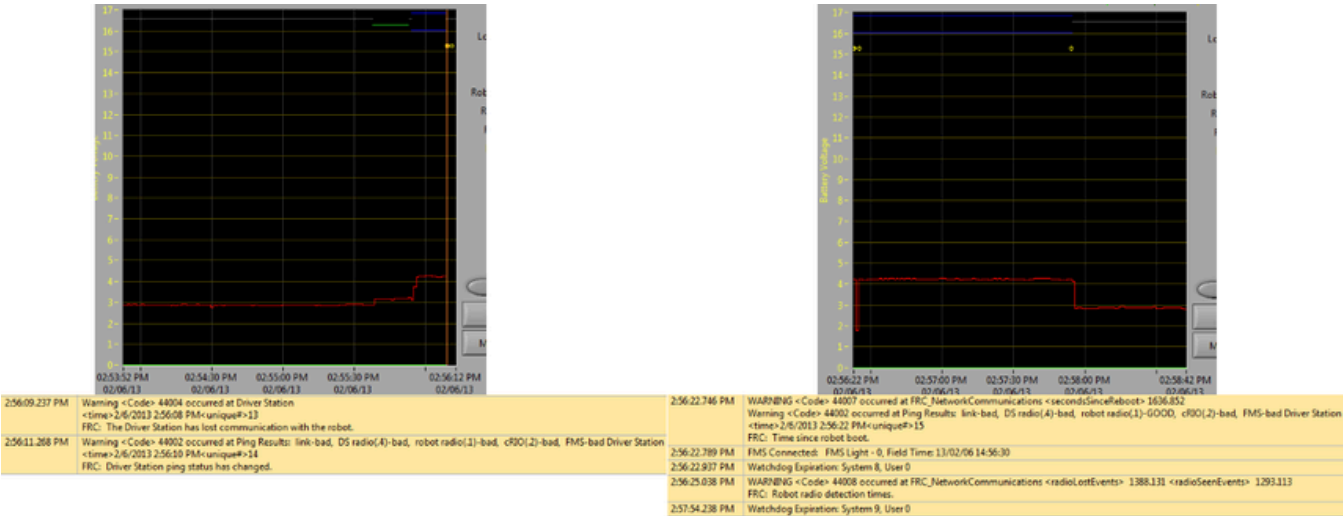
"Normal" Log / “正常”Log



This is an example of a normal match log. The errors and warnings contained in the first box are from when the DS first started and can be ignored. This is confirmed by observing that these events occurred prior to the "FMS Connected:" event. The last event shown can also be ignored, it is also from the robot first connecting to the DS (it occurs 3 seconds after connecting to FMS) and occurs roughly 30 seconds before the match started.

这是正常比赛 Log 的一个示例。第一个输入框中的 Error 和 Warning 来自 DS 初次启动时，可以忽略。可以通过观察这些 Event 发生在“FMS Connected:”Event 之前来确认这一点。显示的最后一个 Event 也可以忽略，它同样来自机器人首次连接到 DS 时；该 Event 发生在连接到 FMS 后 3 秒，约在比赛开始前 30 秒。

Disconnected from FMS / 与 FMS 断开连接



When the DS disconnects from FMS, and therefore the robot, during the match it may segment the log into pieces. The key indicators to this failure are the last event of the first log, indicating that the connection to FMS is now "bad" and the second event from the 2nd log which is a new FMS connected message followed by the DS immediately transitioning into Teleop Enabled. The most common cause of this type of failure is an ethernet cable with no latching tab or a damaged ethernet port on the DS computer.

比赛期间 DS 与 FMS 断开连接，并因此与机器人断开连接时，Log 可能会被分成多个片段。此故障的关键指示包括：第一段 Log 的最后一个 Event，表示与 FMS 的连接现在为“bad”；第二段 Log 的第二个 Event，是一条新的 FMS Connected 消息，随后 DS 立即转换到 Teleop Enabled。此类故障最常见的原因是 Ethernet Cable 没有锁扣，或 DS 计算机上的 Ethernet Port 损坏。

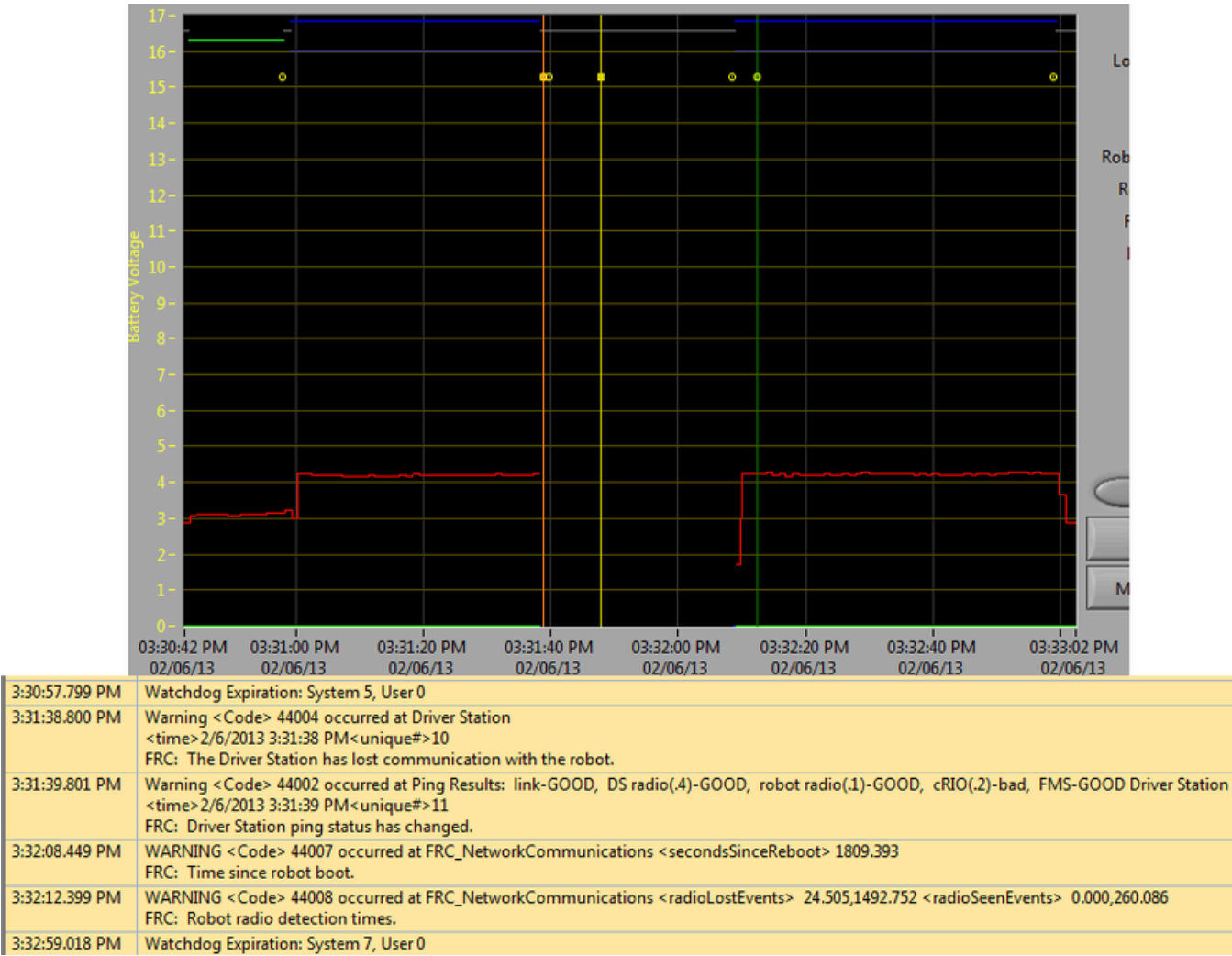
roboRIO Reboot / roboRIO 重启



The "Time since robot boot" message is the primary indicator in a connection failure caused by the roboRIO rebooting. In this log the DS loses connection with the roboRIO at 3:01:36 as indicated by the first event. The second event indicates that the ping initiated after the connection failed was successful to all devices other than the roboRIO. At 3:01:47 the roboRIO begins responding to pings again, one additional ping fails at 3:01:52. At 3:02:02 the Driver Station connects to the roboRIO and the roboRIO reports that it has been up for 3.682 seconds. This is a clear indicator that the roboRIO has rebooted. The code continues to load and at 3:02:24 the code reports an error communicating with the camera. A warning is also reported indicating that no robot code is running right before the code finishes starting up.

“Time since robot boot”消息是判断连接故障由 roboRIO 重启造成的主要依据。在此 Log 中，第一个 Event 表明 DS 于 3:01:36 与 roboRIO 断开连接。第二个 Event 表明，连接失败后发起的 Ping 对除 roboRIO 外的所有设备都成功。roboRIO 于 3:01:47 再次开始响应 Ping，3:01:52 又有一次 Ping 失败。3:02:02，Driver Station 连接到 roboRIO，而 roboRIO 报告其运行时间为 3.682 秒。这清楚地表明 roboRIO 已经重启。随后代码继续加载；3:02:24，代码报告与 Camera 通信时出错。在代码完成启动前，还报告了一条表示没有 Robot Code 正在运行的 Warning。

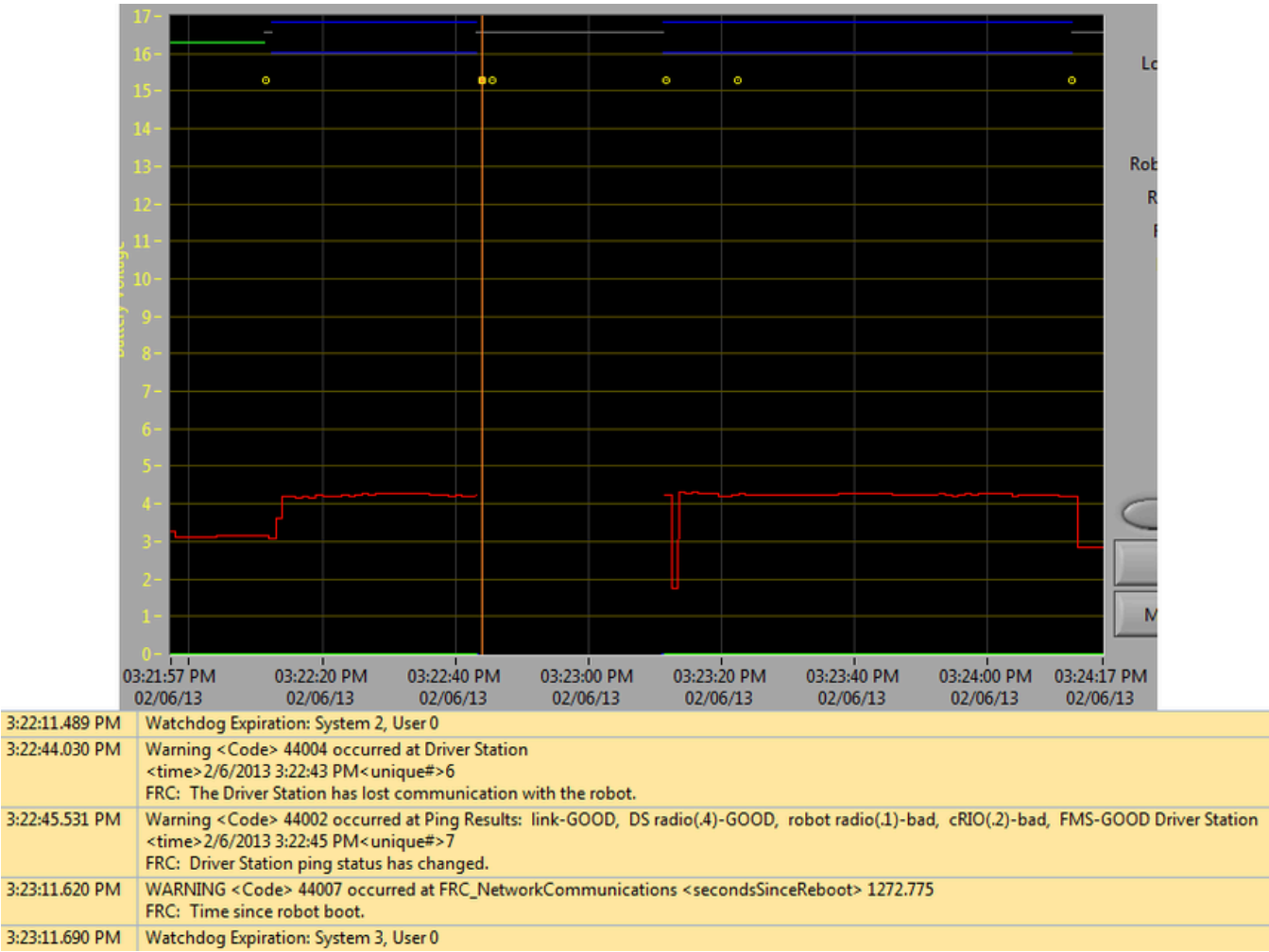
Ethernet cable issue on robot / 机器人上的 Ethernet Cable 问题



An issue with the ethernet cable on the robot is primarily indicated by the ping to the roboRIO going to bad and Radio Lost and Radio Seen events when the roboRIO reconnects. The "Time since robot boot" message when the roboRIO reconnects will also indicate that the roboRIO has not rebooted. In this example, the robot Ethernet cable was disconnected at 3:31:38. The ping status indicates that the radio is still connected. When the robot reconnects at 3:32:08 the "Tim since robot boot" is 1809 seconds indicating that the roboRIO clearly did not reboot. At 3:32:12 the robot indicates that it lost the radio 24.505 seconds ago and it returned 0.000 seconds ago. These points are plotted as vertical lines on the graph, yellow for radio lost and green for radio seen. Note that the times are slightly offset from the actual events as shown via the disconnection and connection, but help to provide additional information about what is occurring.

机器人上的 Ethernet Cable 出现问题时，主要表现为对 roboRIO 的 Ping 变为 bad，并在 roboRIO 重新连接时出现 Radio Lost 和 Radio Seen Event。roboRIO 重新连接时的“Time since robot boot”消息也会表明 roboRIO 并未重启。在此示例中，机器人的 Ethernet Cable 于 3:31:38 断开。Ping Status 表明 Radio 仍保持连接。机器人于 3:32:08 重新连接时，“Tim since robot boot”为 1809 秒，清楚地表明 roboRIO 没有重启。3:32:12，机器人表示它在 24.505 秒前丢失 Radio，并在 0.000 秒前恢复连接。这些时间点以竖线绘制在图表中：黄色表示 Radio Lost，绿色表示 Radio Seen。请注意，根据断开和连接 Event 所示，这些时间与实际 Event 略有偏移，但仍有助于提供有关实际情况的其他信息。

Radio reboot / Radio 重启



A reboot of the robot radio is typically characterized by a loss of connection to the radio for ~40-50 seconds. In this example, the radio briefly lost power at 3:22:44, causing it to start rebooting. The event at 3:22:45 indicates that the ping to the radio failed. At 3:23:11, the DS regains communication with the roboRIO and the roboRIO indicates it has been up for 1272.775 seconds, ruling out a roboRIO reboot. Note that the network switch on the radio comes back up very quickly so a momentary power loss may not result in a "radio lost"/"radio seen" event pair. A longer disturbance may result in radio events being logged by the DS. In that case, the distinguishing factor which points towards a radio reboot is the ping status of the radio from the DS. If the radio resets, the radio will be unreachable. If the issue is a cabling or connection issue on the robot, the radio ping should remain "GOOD".

Robot Radio 重启的典型特征是与 Radio 的连接丢失约 40-50 秒。在此示例中，Radio 于 3:22:44 短暂断电并开始重启。3:22:45 的 Event 表明对 Radio 的 Ping 失败。3:23:11，DS 恢复与 roboRIO 的通信，而 roboRIO 表示其已运行 1272.775 秒，因此可以排除 roboRIO 重启。请注意，Radio 上的 Network Switch 恢复速度很快，因此瞬时断电不一定会产生一对“radio lost”/“radio seen”Event。持续时间更长的干扰可能使 DS 记录 Radio Event。在这种情况下，判断 Radio 是否重启的关键依据是 DS 对 Radio 的 Ping Status。如果 Radio 重置，它将无法访问；如果问题来自机器人上的线缆或连接，Radio Ping 应保持“GOOD”。

Driver Station Errors/Warnings / Driver Station 错误与警告

In an effort to provide both Teams and Volunteers (FTA / CSA / etc.) more information to use when diagnosing robot problems, a number of Warning and Error messages have been added to the Driver Station. These messages are displayed in the DS diagnostics tab when they occur and are also included in the DS Log Files that can be viewed with the Log File Viewer. This document discusses the messages produced by the DS (messages produced by WPILib can also appear in this box and the DS Logs).

为了向队伍和志愿者（FTA / CSA 等）提供更多可用于诊断机器人问题的信息，Driver Station 中增加了多种 Warning 和 Error 消息。这些消息出现时会显示在 DS 的 Diagnostics Tab 中，也会记录在可通过 Log File Viewer 查看的 DS Log File 中。本文档介绍由 DS 生成的消息；WPILib 生成的消息也可能出现在此消息框和 DS Log 中。

Joystick Unplugged / Joystick 已断开

```
ERROR<Code>-44009 occurred at Driver Station
<time>2/5/2013 4:43:54 PM <unique#>1
FRC: A joystick was disconnected while the robot was enabled.
```

This error is triggered when a Joystick is unplugged. Contrary to the message text this error will be printed even if the robot is not enabled, or even connected to the DS. You will see a single instance of this message occur each time the Driver Station is started, even if Joysticks are properly connected and functioning.

拔出 Joystick 时会触发此错误。与消息文字所述不同，即使机器人未启用，甚至尚未连接到 DS，也会输出此错误。每次启动 Driver Station 时都会出现一次该消息，即使 Joystick 已正确连接并正常工作也是如此。

NOTE: Joystick Unplugged warnings can be silenced by calling

`DriverStation.silenceJoystickConnectionWarning(true)` ([Java](#))

说明：可以调用 `DriverStation.silenceJoystickConnectionWarning(true)` ([Java](#)) 屏蔽 Joystick Unplugged 警告。

Lost Communication / 通信丢失

```
Warning<Code>44004 occurred at Driver Station
<time>2/6/2013 11:07:53 AM<unique#>2
FRC: The Driver Station has lost communication with the robot.
```

This Warning message is printed whenever the Driver Station loses communication with the robot (Communications indicator changing from green to red). A single instance of this message is printed when the DS starts up, before communication is established.

每当 Driver Station 与机器人之间的通信丢失时（Communications 指示器由绿色变为红色），都会输出此 Warning 消息。DS 启动后、建立通信前也会输出一此消息。

Ping Status / Ping 状态

```
Warning<Code>44002 occurred at Ping Results: link-GOOD, DS radio(.4)-bad, robot radio(.1)-GOOD, cRIO(.2)-bad, FMS- bad Driver Station
<time>2/6/2013 11:07:59 AM<unique#>5
FRC: Driver Station ping status has changed.
```

A Ping Status warning is generated each time the Ping Status to a device changes while the DS is not in communication with the roboRIO. As communications is being established when the DS starts up, a few of these warnings will appear as the Ethernet link comes up, then the connection to the robot radio, then the roboRIO (with FMS mixed in if applicable). If communications are later lost, the ping status change may help identify at which component the communication chain broke.

当 DS 未与 roboRIO 通信时，只要某个设备的 Ping Status 发生变化，就会生成一条 Ping Status 警告。DS 启动并逐步建立通信时，随着 Ethernet Link 启动、建立与 Robot Radio 的连接、再建立与 roboRIO 的连接，会出现若干此类警告；如果适用，其中还会包含 FMS。之后如果通信丢失，Ping Status 的变化可能有助于判断通信链在哪个组件处中断。

Time Since Robot Boot / 机器人启动后的时间

```
WARNING<Code>44007 occurred at FRC_NetworkCommunications
**<secondsSinceReboot> 3.585**
FRC: Time since robot boot.
```

This message is printed each time the DS begins communicating with the roboRIO. The message indicates the up-time, in seconds, of the roboRIO and can be used to determine if a loss of communication was due to a roboRIO Reboot.

每当 DS 开始与 roboRIO 通信时，都会输出此消息。该消息以秒为单位表示 roboRIO 的运行时间，可用于判断通信丢失是否由 roboRIO 重启导致。

Radio Detection Times / Radio 检测时间

```
WARNING<Code>44008 occurred at FRC_NetworkCommunications
<radioLostEvents> 19.004<radioSeenEvents> 0.000
FRC: Robot radio detection times
WARNING<Code>44008 occurred at FRC_NetworkCommunications
<radioLostEvents> 2.501,422.008<radioSeenEvents> 0.000,147.005
FRC: Robot radio detection times.
```

This message may be printed when the DS begins communicating with the roboRIO and indicates the time, in seconds, since the last time the radio was lost and seen. In the first example image above the message indicates that the roboRIO's connection to the radio was lost 19 seconds before the message was printed and the radio was seen again right when the message was printed. If multiple radioLost or radioSeen events have occurred since the roboRIO booted, up to 2 events of each type will be included, separated by commas.

DS 开始与 roboRIO 通信时可能会输出此消息。该消息以秒为单位，表示距离上一次丢失 Radio 和重新检测到 Radio 所经过的时间。在上方第一个示例中，该消息表示 roboRIO 与 Radio 的连接在消息输出前 19 秒丢失，并在消息输出时重新检测到 Radio。如果 roboRIO 启动后发生过多 `radioLost` 或 `radioSeen` 事件，则每种类型最多会包含 2 个事件，并以逗号分隔。

No Robot Code / 没有 Robot Code

```
Warning<Code>44003 occurred at Driver Station
<time>2/8/2013 9:50:13 AM<unique#>8
FRC: No robot code is currently running.
```

This message is printed when the DS begins communicating with the roboRIO, but detects no robot code running. A single instance of this message will be printed if the Driver Station is open and running while the roboRIO is booting as the DS will begin communication with the roboRIO before the robot code finishes loading.

当 DS 开始与 roboRIO 通信、但检测到没有 Robot Code 正在运行时，会输出此消息。如果 roboRIO 启动时 Driver Station 已打开并运行，则会输出一次该消息，因为 DS 会在 Robot Code 完成加载前开始与 roboRIO 通信。

Driver Station Timing Viewer / Driver Station Timing Viewer

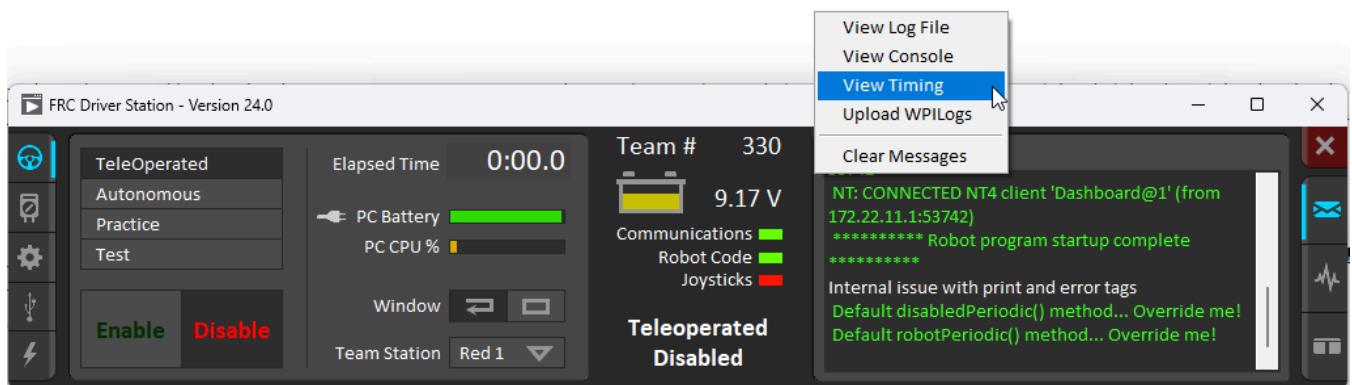
The 2024 Driver Station has a new window to help diagnose robot control issues.

2024 Driver Station 新增了一个窗口，用于协助诊断机器人控制问题。

Opening the Timing Viewer / 打开 Timing Viewer

To start the Driver Station Timing Viewer, select the gear icon and then select **View Timing**.

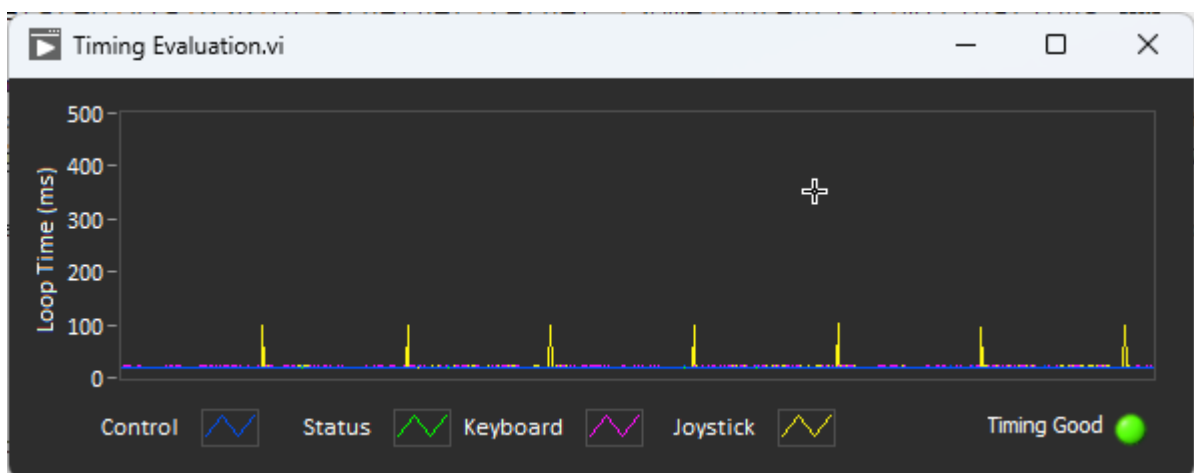
要启动 Driver Station Timing Viewer，请选择齿轮图标，然后选择 **View Timing**。



Viewing Timing / 查看 Timing

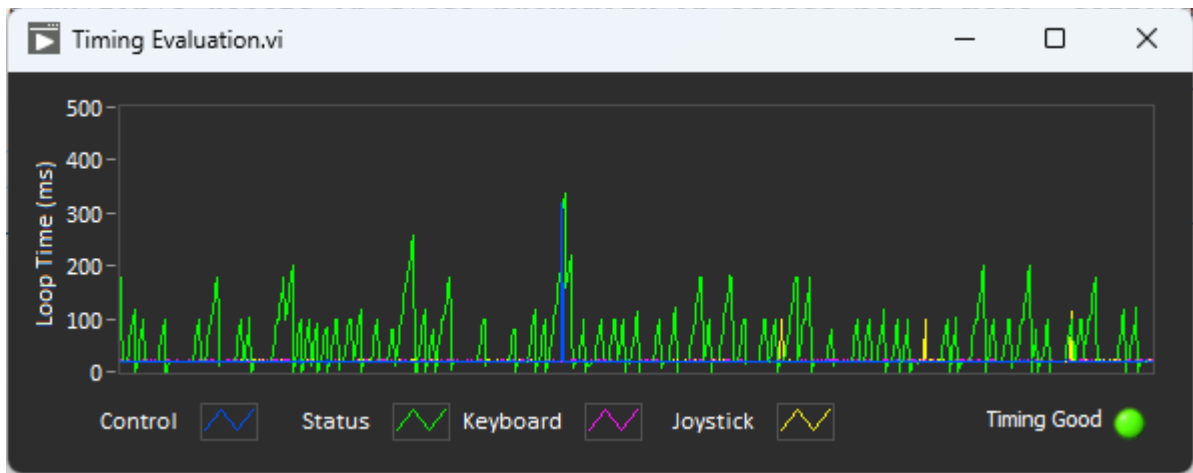
The Timing viewer shows the timing of the driver station loops measuring the joystick, keyboard, and control, and status network packets. When timing is good, all values should be close to 0 ms. This can help diagnose what is causing robot control issues.

Timing Viewer 显示 Driver Station 各循环的时序，包括对 Joystick、Keyboard、Control Network Packet 和 Status Network Packet 的测量。时序正常时，所有数值都应接近 0 ms。这有助于诊断导致机器人控制问题的原因。



The next image shows what it looks like when network congestion causes network packets to be delayed and combined. In this example, the communication was so bad that the robot wouldn't stay enabled or connected for more than a second.

下一张图片展示了网络拥塞导致 Network Packet 延迟并合并时的情况。在此示例中，通信状况非常糟糕，机器人无法保持启用或连接超过一秒。



Manually Setting the Driver Station to Start Custom Dashboard / 手动设置 Driver Station 以启动自定义 Dashboard

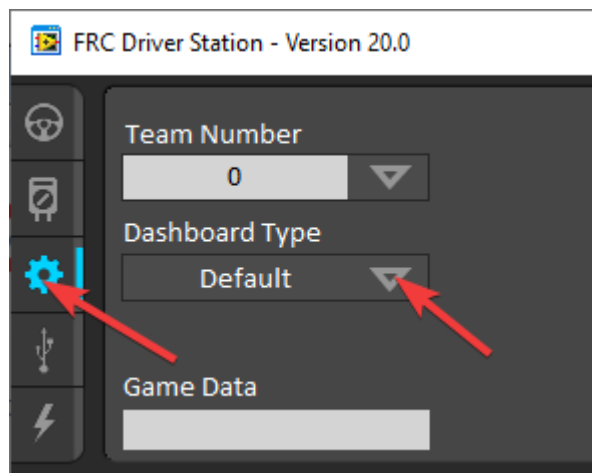
NOTE: If WPILib is not installed to the default location (such as when files are copied to a PC manually), the dashboard of choice may not launch properly. To have the DS start a custom dashboard when it starts up, you have to manually modify the settings for the default dashboard.

说明: 如果 WPILib 未安装到默认位置（例如手动将文件复制到 PC 时），所选 Dashboard 可能无法正常启动。若要让 DS 在启动时打开自定义 Dashboard，必须手动修改默认 Dashboard 的设置。

WARNING: This is not needed for most installations, try using the appropriate [Dashboard Type setting](#) for your language first.

警告: 大多数安装无需执行此操作。请先尝试使用适合所用语言的 [Dashboard Type 设置](#)。

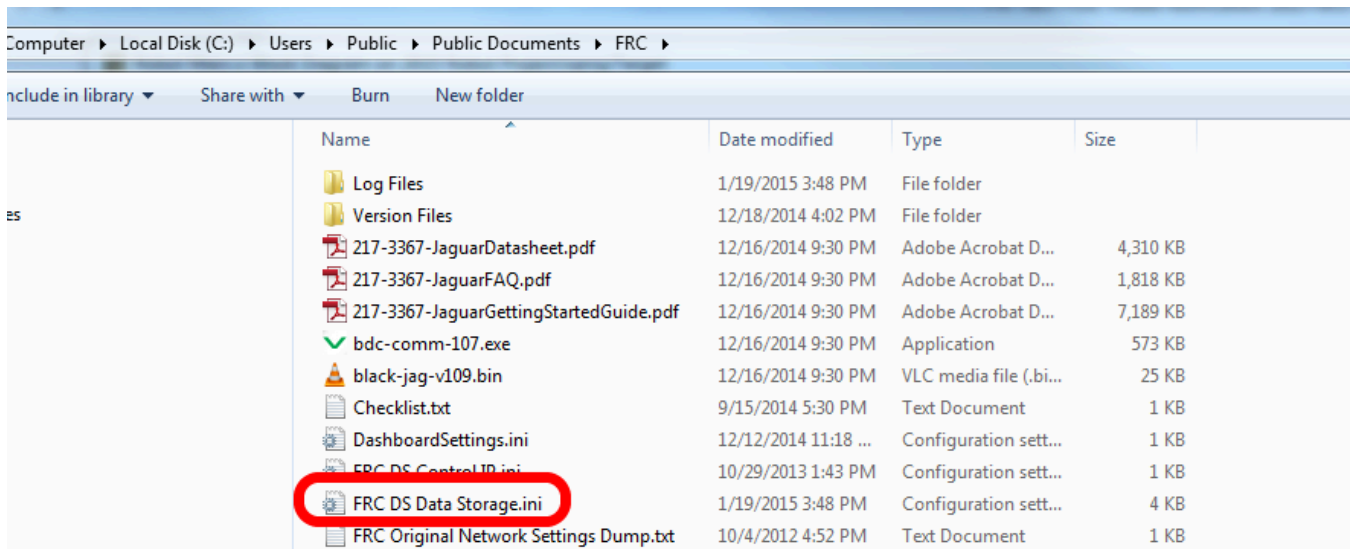
Set Driver Station to Default / 将 Driver Station 设置为 Default



Open the Driver Station software, click on the Setup tab and set the Dashboard setting to Default. **Then close the Driver Station!**

打开 Driver Station 软件，单击 Setup Tab，并将 Dashboard 设置为 Default。然后关闭 **Driver Station!**

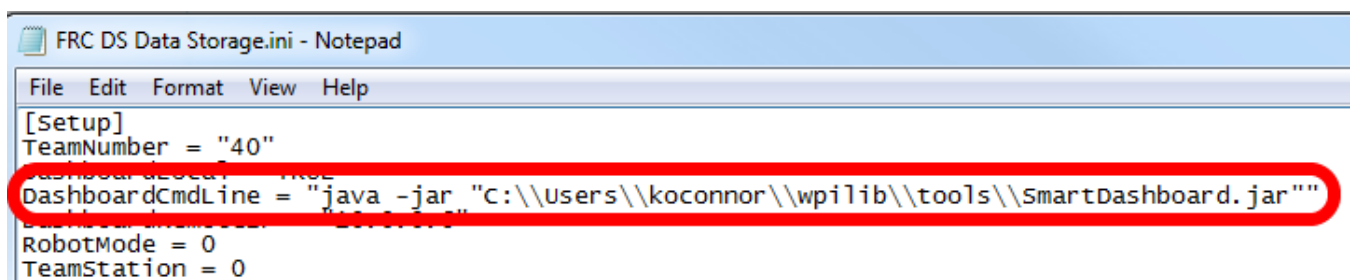
Open DS Data Storage file / 打开 DS Data Storage 文件



Browse to `C:\Users\Public\Documents\FRC` and double click on `FRC DS Data Storage` to open it.

浏览到 `C:\Users\Public\Documents\FRC`，然后双击 `FRC DS Data Storage` 将其打开。

DashboardCmdLine / DashboardCmdLine



Locate the line beginning with `DashboardCmdLine`. Modify it to point to the dashboard to launch when the driver station starts

找到以 `DashboardCmdLine` 开头的行。修改该行，使其指向 Driver Station 启动时要打开的 Dashboard。

Java Dashboard / Java Dashboard

Replace the string after `=` with `java -jar "C:\PATH\TO\DASHBOARD.jar"` where the path specified is the path to the dashboard `jar` file. Save the `FRC DS Data Storage` file.

将 `=` 后的字符串替换为 `java -jar "C:\PATH\TO\DASHBOARD.jar"`，其中指定路径是 Dashboard `jar` 文件的路径。保存 `FRC DS Data Storage` 文件。

TIP: Shuffleboard and Smartdashboard require Java 17.

提示: Shuffleboard 和 Smartdashboard 需要 Java 17。

Dashboard from WPILib installer / WPILib Installer 提供的 Dashboard

Replace the string after `=` with `C:\Users\Public\wpilib\YYYY\tools\DASHBOARD.exe` where `YYYY` is the year and `DASHBOARD.exe` is `Shuffleboard.exe`, `Smartdashboard.exe`, or `Elastic.exe`. For example `DashboardCmdLine = "wscript C:\Users\Public\wpilib\2026\tools\Elastic.exe"`. Save the `FRC DS Data Storage` file.

将 `=` 后的字符串替换为 `C:\Users\Public\wpilib\YYYY\tools\DASHBOARD.exe`，其中 `YYYY` 表示年份，`DASHBOARD.exe` 为 `Shuffleboard.exe`、`Smartdashboard.exe` 或 `Elastic.exe`。例如：`DashboardCmdLine = "wscript C:\Users\Public\wpilib\2026\tools\Elastic.exe"`。保存 `FRC DS Data Storage` 文件。

Launch Driver Station / 启动 Driver Station

The Driver Station should now launch the dashboard each time it is opened.

现在，每次打开 Driver Station 时，它都应启动该 Dashboard。